

PMATH 347: Groups and Rings with Professor David McKinnon

Introduction to Groups

A group is a bunch of things you can multiply and divide (or add and subtract) in a sensible way.

Examples:

The real numbers $\mathbb{R}$ (add and subtract) and the nonzero real numbers $\mathbb{R}^*$ (multiply and divide), similarly for $\mathbb{C}$ and $\mathbb{C}^*$.

$\text{GL}(n, \mathbb{R})$, invertible $n \times n$ matrices with real entries.

$S^1 = \{z \in \mathbb{C} \mid |z| = 1\}$, the unit circle of complex numbers under multiplication.

$S_n := \{f : \{1, \dots, n\} \rightarrow \{1, \dots, n\} \mid f \text{ is bijective}\}$.

This is the formal definition, which is necessary to cover, but not where your intuition should go (in the words of McKinnon, “stupid but necessary”).

Definition:

A group is an ordered pair $(G, \cdot)$ where G is a set and $\cdot : G \times G \rightarrow G$ is a function satisfying:

- (1): If $a, b, c \in G$, then $(a \cdot b) \cdot c = a \cdot (b \cdot c)$
- (2): There is $1 \in G$ such that if $a \in G$, then $1 \cdot a = a \cdot 1 = a$.
- (3): If $a \in G$, then there is $a^{-1} \in G$ such that $a \cdot a^{-1} = a^{-1} \cdot a = 1$.

Definition:

A subgroup of a group G is a subset $H \subseteq G$ that is also a group with the same operation as G . This is sometimes denoted $H \leq G$.

Examples:

We have $S^1 \leq \mathbb{C}^*$ and $\mathbb{R}^* \leq \mathbb{C}^*$.

Another important example is $\text{SL}(n, \mathbb{R})$, the group $n \times n$ matrices with determinant 1. This is a subgroup of $\text{GL}(n, \mathbb{R})$.

Theorem (Subgroup Theorem):

Let G be a group, and $H \subseteq G$ a nonempty subset. Then H is a subgroup of G if and only if for every $a, b \in H$, we have $ab \in H$ and $a^{-1} \in H$.

Proof:

($\implies$) This is true because H is a group.

($\impliedby$) We assume $ab \in H$ and $a^{-1} \in H$ for all $a, b \in H$. We want to show that H is a subgroup of G . The fact that $ab \in H$ means that $\cdot : H \times H \rightarrow H$ is well defined. So it makes sense to check (1), (2), (3) from above:

(1) True because H 's elements are in G and $\cdot$ is G 's operation restricted to H , so the same property holds.

(2) We need to check that $1 \in H$. Choose $a \in H$. Then $1 = a \cdot a^{-1} \in H$ because $a, a^{-1} \in H$ and $\cdot$ is well-defined.

(3) holds for the same reason as (2). ■

Example:

Take $\text{GL}(n, \mathbb{R})$ as the main group. We have seen $\text{SL}(n, \mathbb{R})$ is a subgroup, but so too is the trivial subgroup $\{1\}$ (this works for every group), which in this case is the $n \times n$ identity matrix. We can see this is a subgroup by the above theorem. Another subgroup of $\text{GL}(n, \mathbb{R})$ is $\mathcal{O}(n, \mathbb{R})$, which is the set of $n \times n$ matrices M such that for all $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$, $|\mathbf{v} - \mathbf{w}| = |M\mathbf{v} - M\mathbf{w}|$ (it preserves distances). This is a common way to define subgroups; take a property and then take all the things that don't change this property.

As a general principle, a subset S of a group G that is all the elements that don't change "\$thing", then it's probably a subgroup.

Question:

Is $\mathbb{Z}/7\mathbb{Z}$ a subgroup of $\mathbb{Z}$?

Answer:

No. Because it's not even a subset, $2 \pmod{7}$ is NOT 2, it's the equivalence class. They have the same names, but they are different.

Example:

Take a regular hexagon H , and take the group

$$D_6 = \{ 2 \times 2 \text{ invertible matrices that map } H \rightarrow H \}$$

(this works for any regular n -gon). What matrices are these? Assume that $(1,0)$ is a vertex of H . If $M \in D_6$, then $M(1,0)$ is one of the six vertices of H , so there are six possibilities for the first column of M . Take an adjacent vertex to $(1,0)$. M 's action on it must be a vertex adjacent to $(1,0)$ for the edge to be preserved, giving 2 choices. So there are at most 12 matrices in D_6 . There are actually 12 – 6 rotations, and the choice for the adjacent vertex can either be maintained or reflected after. Similarly, D_n has $2n$ elements, calculated the same way.

Example:

Now let's look at S_n , the symmetric group on n letters, equivalently, the permutations of $\{1, \dots, n\}$, equivalently, bijections $f : \{1, \dots, n\} \rightarrow \{1, \dots, n\}$. There is a helpful notation to discuss this group, called disjoint cycle notation. It works like this. Say we have a permutation σ of $\{1, \dots, n\}$. Let us first ask where 1 goes under the action of σ . Say 1 goes to 2. Then write 12. Now where does 2 go? to 5. So write 125. Where does 5 go? to 4. Then write 1254. Where does 4 go? to 1. So write (1254). 3 is left on its own, so write (1254)(3), occasionally we omit the singleton. Writing a permutation in terms of disjoint cycles like this always works.

Definition:

The order of an element $g \in G$ is the smallest positive integer n satisfying $g^n = 1$. If there is no such integer, then g has infinite order.

Definition:

The order of a group G is its cardinality as a set.

Example:

The order of S_n is $n!$, the order of D_n is $2n$. In disjoint cycle notation for S_n , the order is easily readable: it is the lowest common multiple of the length of the cycles.

Example:

The final example to consider is the Quaternion group Q , defined as the set of matrices:

$$\left\{ \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix} \right\}$$

With the appropriate labels, you get $I^2 = J^2 = K^2 = -1, IJ = K, JK = I, KI = J$.

Definition:

A group G is abelian if and only if $ab = ba$ for every $a, b \in G$.

Definition:

The direct product of groups G and H is the group $G \times H = \{(g, h) \mid g \in G, h \in H\}$ with $(g, h)(g', h') = (gg', hh')$.

Definition:

Let G and H be groups. A homomorphism from G to H is a function $f : G \rightarrow H$ satisfying $f(ab) = f(a)f(b)$ for all $a, b \in G$. One can see that homomorphisms preserve the identity and group inverses.

Definition:

An isomorphism from G to H is a homomorphism $f : G \rightarrow H$ with an inverse homomorphism $f^{-1} : H \rightarrow G$. If there is an isomorphism between G and H , then we say G and H are isomorphic and write $G \cong H$.

Theorem:

A homomorphism $f : G \rightarrow H$ is an isomorphism if and only if it is bijective.

Proof:

($\implies$): f has an inverse (it is an isomorphism), thus it is bijective.

($\impliedby$): Let $f : G \rightarrow H$ be a bijective homomorphism. Since f is bijective, it has a unique inverse $f^{-1} : H \rightarrow G$. We need to show it is a homomorphism. So let $a, b \in H$. We want to show

$$f^{-1}(a)f^{-1}(b) = f^{-1}(ab)$$

It's enough to show

$$f(f^{-1}(ab)) = f(f^{-1}(a)f^{-1}(b))$$

because f is injective. This is just $ab = f(f^{-1}(a)f^{-1}(b))$ so we compute:

$$f(f^{-1}(a)f^{-1}(b)) = ff^{-1}(a)ff^{-1}(b) = ab \quad \blacksquare$$

Examples:

$\det : GL_n(\mathbb{R}) \rightarrow \mathbb{R}^*$ is a homomorphism since $\det(AB) = \det A \det B$, but not an isomorphism since it's not injective.

$q : \mathbb{Z} \rightarrow \mathbb{Z}/7\mathbb{Z}$ which is $q(n) = n \pmod{7}$ because $q(m+n) = q(m) + q(n)$ is also a homomorphism but not an isomorphism.

Define a homomorphism $i : S_n \rightarrow S_{n+1}$ as follows. Define $i(\sigma) : \{1, \dots, n+1\} \rightarrow \{1, \dots, n+1\}$ by

$$[i(\sigma)](k) = \begin{cases} \sigma(k) & k \in \{1, \dots, n\} \\ k & \text{else} \end{cases}. \text{ This is a homomorphism but not an isomorphism. To prove the}$$

former, we need to show $i(\sigma \circ \tau) = i(\sigma) \circ i(\tau)$. This turns out to be straightforward but confusing and annoying.

$\log : \mathbb{R}_{>0} \rightarrow \mathbb{R}$ is a homomorphism, with $\log(xy) = \log(x) + \log(y)$, and an isomorphism (with inverse $\exp$).

The reason we care about homomorphisms is that they preserve algebraic relations: for instance if $a^2 = 1$, $f(a)^2 = f(a)f(a) = f(a^2) = f(1) = 1$.

Example:

$f : G \rightarrow G$ defined by $f(a) = 1$ for all $a \in G$ is the so-called “trivial homomorphism”.

Group Actions

Definition:

An action of a group G on a set S is a homomorphism $\phi : G \rightarrow \text{Sym}(S)$, where $\text{Sym}(S)$ is the group of

bijections from S to itself. Often people write gx to mean $\phi(g)(x)$ for some $g \in G, x \in S$.

Example:

Let $\phi : \text{GL}(n, \mathbb{R}) \rightarrow \text{Sym}(\mathbb{R}^n)$ be the map $M \mapsto (\mathbf{v} \mapsto M\mathbf{v})$. This is an action.

Example:

Let $\phi : \text{GL}(n, \mathbb{R}) \rightarrow \text{Sym}(\mathbb{R}^n)$ be the map $M \mapsto (\mathbf{v} \mapsto (\det M)\mathbf{v})$. This is an action.

Example:

Let $\phi : S_n \rightarrow S_n$ be given by $\phi(\sigma) = \sigma$ (sometimes called the “permutation action”). This is an action of S_n on $\{1, \dots, n\}$.

Example:

The trivial action is $\phi : G \rightarrow \text{Sym}(S)$, given by $\phi(g) = \text{Id}$ for all $g \in G$.

Example:

Let $\phi : D_n \rightarrow \text{Sym}(n\text{-gon})$ be given by $\phi(\sigma) = \sigma$. This is an action of D_n on a regular n -gon.

Definition:

An action ϕ from G to $\text{Sym}(S)$ is

- (1) free if and only if $\phi(g)$ fixes no element of S unless $g = 1$.
- (2) faithful if and only if ϕ is injective. That is, every non-identity element of g moves something in S .
- (3) transitive if and only if for every $x, y \in S$ there is $g \in G$ such that $\phi(g)(x) = y$.

Note that every free action is faithful.

Definition:

Let $\phi : G \rightarrow \text{Sym}(S)$ be an action and let $x \in S$. The orbit of x is $\mathcal{O}_x = \{gx \mid g \in G\} \subseteq S$, that is, the set of all things x can be moved to by the action of G . Notice that if $x \in \mathcal{O}_y$ then

$$gy = x \implies y = g^{-1}x \implies y \in \mathcal{O}_x$$

and since $x \in \mathcal{O}_x$ (take $g = 1$) we see that the orbits $\mathcal{O}_x$ partition S .

Definition:

Let $x \in S$. The stabilizer of x is $\text{Stab}(x) = \{g \in G \mid gx = x\}$.

Example:

Let G be a group. Then G acts on itself by left multiplication: take $\phi : G \rightarrow \text{Sym}(G)$, where $\phi(g) = (x \mapsto gx)$. This is an action: we need to check $\phi(gh) = \phi(g)\phi(h)$, which we can see is true:

$$(\phi(g)\phi(h))(x) = (\phi(g))(\phi(h)(x)) = (\phi(g))hx = ghx$$

This free, and therefore automatically also faithful. It's also transitive. For any x , $\text{Stab}(x) = 1$ and $\mathcal{O}_x = G$. Say G is a finite group. So we can number the elements of G as $\{1, \dots, n\}$, so $\text{Sym}(G) \cong \text{Sym}\{1, \dots, n\} = S_n$. So this action gives an injective homomorphism $G \hookrightarrow S_n$ (because it is faithful). In particular, every finite group is isomorphic to a subgroup of S_n , where $n = |G|$. This is called **Cayley's Theorem**.

Example:

Consider the action $\phi : G \rightarrow \text{Sym}(G)$ given by $\phi(g) = (x \mapsto g^{-1}xg)$. This is called conjugation. It's never free (unless the group is trivial) since $gxg^{-1} = x \iff gx = xg$ which happens all the time ($x = g$). It's sometimes faithful, sometimes not, and it is not transitive since 1 is fixed by every element of G . For any x , $\text{Stab}(x)$ (all g such that $gx = xg$) is called the centralizer of x . The orbit $\mathcal{O}_x$ of x is complicated, it's called the conjugacy class of x .

Cyclic Groups and Lagrange's Theorem

Definition:

Let G be a group, $g \in G$ be any element. The subgroup generated by g is $\langle g \rangle = \{g^n \mid n \in \mathbb{Z}\} = \{\dots, g^{-2}, g^{-1}, 1, g, g^2, \dots\}$. Since $\langle g \rangle$ is closed under multiplication and inversion, it is a subgroup.

Definition:

A cyclic group is a group of the form $\langle g \rangle$ for some g .

Example:

Take $G = \mathbb{Z}$, $g = 1$. Then $\langle g \rangle = \{\dots, -2, -1, 0, 1, 2, \dots\} = \mathbb{Z}$. So $\mathbb{Z}$ is a cyclic group.

Definition

Let G be a group, $S \subseteq G$. The smallest subgroup of G that contains S is $\langle S \rangle = \bigcap_{S \subseteq H \leq G} H$

Example:

Take $G = \mathbb{Z}/n\mathbb{Z}$, $g = 1$. Then $\langle g \rangle = \{0, 1, 2, \dots, n-1\} = \mathbb{Z}/n\mathbb{Z}$, so $\mathbb{Z}/n\mathbb{Z}$ is a cyclic group.

Theorem:

Let $g \in G$, $g \in G$ any element. Then if g has infinite order, $\langle g \rangle \cong \mathbb{Z}$, and if g has finite order n , $\langle g \rangle \cong \mathbb{Z}/n\mathbb{Z}$.

Proof:

Define a function $\phi : \mathbb{Z} \rightarrow \langle ga \rangle$ by $\phi(n) = g^n$. By the rules of exponents, this is a homomorphism. ϕ is clearly surjective, because we hit every integer power of g . If g has infinite order, then ϕ is injective:

$$\phi(n) = \phi(m) \iff g^n = g^m \iff g^{n-m} = 1 \iff n = m$$

so we are done in this case because we just wrote down an isomorphism, so $\mathbb{Z} \cong \langle g \rangle$. If g has finite order n , then ϕ is well defined mod n , i.e. if $k \sim l \pmod{n}$, then $\phi(k) = g^k = g^l = \phi(l)$. So if g has order n , we can define $\bar{\phi} : \mathbb{Z}/n\mathbb{Z} \rightarrow \langle g \rangle$ by $\bar{\phi}(k \pmod{n}) = \phi(k) = g^k$. This is surjective because ϕ is, and $\langle g \rangle$ has n elements $\{1, g, g^2, \dots, g^{n-1}\}$. So $\bar{\phi}$ is a surjective function of a set from n elements to a set of n elements, and thus $\bar{\phi}$ is injective. It's a homomorphism because ϕ is, therefore it is an isomorphism. ■

Theorem (Lagrange)

Let G be a group, $H \leq G$ a subgroup. Then $|G|$ is divisible by $|H|$.

Proof:

There is an action of H on G by left multiplication:

$$\phi : H \rightarrow \text{Sym}(G), \phi(h)(g) = hg$$

The orbit of 1 under this action is H . The orbit of g is $Hg = \{hg \mid h \in H\}$ and there is a bijection $H \rightarrow Hg$, $h \mapsto hg$, with inverse $a \mapsto ag^{-1}$, and we know G is the disjoint union of the orbits, all of which have size $|H|$, so $|G| = |H|(\# \text{ of orbits})$. ■

Definition

Hg is a right coset of H in G . The number of right cosets is the index of H in G , and is written $[G : H]$. If G is finite, then $[G : H] = \frac{|G|}{|H|}$.

Corollary

Let G be a group, $x \in G$ any element. Then $|G|$ is divisible by the order of x .

Proof:

Apply Lagrange's theorem to $\langle g \rangle$. ■

Kernels of Homomorphisms, Normal Subgroups, and Quotients

Say $f : G \rightarrow H$ is a group homomorphism. How do we tell if f is injective?

Definition

The kernel of f is $\ker f = \{g \in G \mid f(g) = 1\}$.

Theorem

A homomorphism f is injective if and only if $\ker f = \{1\}$.

Proof:

The forward direction is clear, since nothing else can go to 1 when f is injective besides the identity.

For the reverse direction, suppose $\ker f = \{1\}$. Then if $f(a) = f(b)$, we get

$$f(a)f(b)^{-1} = 1 \implies f(ab^{-1}) = 1 \implies ab^{-1} \in \ker f$$

Thus $ab^{-1} = 1$, so $a = b$. Thus f is injective. ■

Proposition:

For any homomorphism $f : G \rightarrow H$, $\ker f$ is a subgroup of G .

Proof:

If $f(a) = 1$, then $1 = f(a)^{-1} = f(a^{-1})$, so its closed under inversion.

For products, if $f(a), f(b) = 1$ for $a, b \in G$, then $f(ab) = f(a)f(b) = 1$, so $\ker f$ is closed under products.

Finally it's nonempty since $f(1) = 1$. ■

Example:

Say $a \in \ker f$, $g \in G$. Then $f(gag^{-1}) = f(g)f(a)f(g)^{-1} = 1$, so $gag^{-1} \in \ker f$. In particular, $\ker f$ is closed under conjugation, motivating the definition:

Definition:

A subgroup $H \leq G$ is called normal if and only if for every $h \in H$ and $g \in G$, we have $ghg^{-1} \in H$. This is sometimes written $gHg^{-1} \subseteq H$.

Example:

$SL_n(\mathbb{R})$ is a normal subgroup of $GL_n(\mathbb{R})$ because if $\det A = 1$, then $\det(PAP^{-1}) = 1$. A cheaper way to show this is that $\det : GL_n(\mathbb{R}) \rightarrow \mathbb{R}^*$ has $\ker \det = SL_n(\mathbb{R})$.

Example:

If G is abelian, then every subgroup is normal. The converse is false (the quaternion group is the simplest counterexample).

Example:

Let $G = D_n$, the symmetries of a regular n -gon. Let H be the subgroup of rotations. It's a normal subgroup, because it's the kernel of the homomorphism $\phi : D_n \rightarrow \mathbb{Z}_2$ given by $\phi(\sigma) = \begin{cases} 0 & \text{if rotation} \\ 1 & \text{if reflection} \end{cases}$.

Theorem:

Let H be a subgroup of G . Then H is normal if and only if there is a group P and a homomorphism $\phi : G \rightarrow P$ such that $\ker \phi = H$.

Proof:

($\Leftarrow$): Proved above.

($\Rightarrow$): Say H is normal. We want to define a group P and a homomorphism $\phi : G \rightarrow P$ whose kernel is H . Now, suppose we have such a ϕ . Then it will map H to $\{1\} \subseteq P$. If $g \notin H$, then $\phi(g) \neq 1$ in P . If g' satisfies $\phi(g') = \phi(g)$, what can we say about g' ? Then $\phi(g^{-1}g') = 1$, implying $g^{-1}g' \in H$, implying $g' \in gH$. Therefore, we define $P = \{gH \mid g \in G\}$. This tells us what P is as a set, define

a group multiplication by $(g_1H)(g_2H) = g_1g_2H$. Is this well-defined? Yes, it is. Say $g_1H = g'_1H$ and $g_2H = g'_2H$, we need to check that $g_1g_2H = g'_1g'_2H$.

Lemma

If H is normal in G , then for all $g \in G$, $gH = Hg$.

Proof of lemma:

Say $gh \in gH$. We want to show that $gh \in Hg$. But H is normal, so $ghg^{-1} \in H$ and thus $gh \in Hg$, so $gH \subseteq Hg$, the reverse inclusion is similar. ■

Rest of the proof:

So $g_1g_2H = g_1g'_2H$ by assumption, which equals $g_1Hg'_2 = g'_1Hg'_2 = g'_1g'_2H$ by the lemma applied a few times and our hypotheses. It's straightforward to show that this operation makes P into a group (inheriting from G 's group operation). Now define $\phi : G \rightarrow P$ by $\phi(g) = gH$, which is a homomorphism because the multiplication in P is well-defined. It's also easy to see $\ker \phi = H$, because if $x \in \ker \phi$ then $xH = 1H = H$, implying $x \in H$ (since its inverse is in H), and if $x \in H$ then $\phi(x) = xH = H$ by closure of H 's group operation. ■

Definition

The group P constructed in the above proof is called the quotient of G by H and is written G/H .

Definition

Let $\phi : G \rightarrow H$ be a homomorphism. The image of ϕ is $\text{im } \phi = \{h \in H \mid \exists g \in G \text{ s.t. } h = \phi(g)\}$, which is a subgroup of H .

Theorem (Universal Property of Quotients):

Let G and H be groups, with N a normal subgroup. Suppose there exists a homomorphism $f : G \rightarrow H$, and let $q : G \rightarrow G/N$ be the quotient homomorphism given by reduction mod N . There exists a homomorphism $\tilde{f} : G/N \rightarrow H$ satisfying $f = \tilde{f} \circ q$ if and only if $N \subseteq \ker f$. If such an $\tilde{f}$ exists, $\text{im } \tilde{f} = \text{im } f$ and $\ker \tilde{f} = q(\ker f)$.

Remark:

One should think of $\tilde{f}$ as the reduction “ $f \pmod{N}$ ”.

Example:

Take $GL_n(\mathbb{R})$ mapping to $\mathbb{R}^*$ by the determinant, where we take $N = SL_n(\mathbb{R})$. Since $\ker(\det) = SL_n(\mathbb{R})$, there is a “ $\det$ ” by the UPQ, $\widetilde{\det} : GL_n(\mathbb{R})/SL_n(\mathbb{R}) \rightarrow \mathbb{R}^*$, satisfying $\widetilde{\det} \circ q = \det$. Thus $\text{im } \widetilde{\det} = \text{im } \det = \mathbb{R}^*$, and $\ker \widetilde{\det} \pmod{SL_n(\mathbb{R})} = 1 \pmod{SL_n(\mathbb{R})}$. So, $\widetilde{\det}$ is an isomorphism! Therefore $GL_n(\mathbb{R})/SL_n(\mathbb{R}) \cong \mathbb{R}^*$.

Example:

Let $G = \mathbb{Z}^2$, $N = \langle (1, 2) \rangle$. What is G/N like? Well we have the reduction map q , we want to find a homomorphism $f : \mathbb{Z}^2 \rightarrow H$ such that $\langle (1, 2) \rangle \subseteq \ker f$. Say $H = \mathbb{Z}$, and $f(a, b) = 2a - b$. Then f is a homomorphism. This yields, by the UPQ, a homomorphism $\tilde{f} : \mathbb{Z}^2/\langle (1, 2) \rangle \rightarrow \mathbb{Z}$, with $\tilde{f}((a, b) \pmod{N}) = 2a - b$ because $\langle (1, 2) \rangle \subseteq \ker f$, and in fact this is an equality, so $\ker \tilde{f} = \ker f \pmod{\langle (1, 2) \rangle} = 1 \pmod{\langle (1, 2) \rangle}$, so f is injective. Moreover, since $f(0, -b) = b$, f is onto, so $\tilde{f}$ is onto, so $\tilde{f}$ is an isomorphism, so $\mathbb{Z}^2/\langle (1, 2) \rangle \cong \mathbb{Z}$.

Proof of UPQ:

($\implies$) : We want to show that $N \subseteq \ker f$. If $n \in N$, then $f(n) = \tilde{f}(q(n)) = \tilde{f}(1) = 1$.

($\impliedby$) : Say $N \subseteq \ker f$. Define $\tilde{f} : G/N \rightarrow H$ by $\tilde{f}(gN) = f(g)$. To check that this is well-defined, say $g_1N = g_2N$, then $\tilde{f}(g_1N) = f(g_1)$ and $\tilde{f}(g_2N) = f(g_2)$. But $g_2^{-1}g_1 \in N$, so $f(g_2^{-1}g_1) = 1$, implying

$f(g_1) = f(g_2)$. This completes the proof of this direction.

The facts that $\text{im } \tilde{f} = \text{im } f$ and $\ker \tilde{f} = \ker f \pmod{N}$ follow immediately from $f = \tilde{f} \circ q$ and the surjectivity of q . ■

The following are consequences of the UPQ:

Theorem (First Isomorphism Theorem):

Let $f : G \rightarrow H$ be a homomorphism. Then $\text{im } f \cong G / \ker f$.

Proof:

The Universal Property of Quotients tells us that there is $\tilde{f} : G / \ker f \rightarrow H$. The kernel of $\tilde{f}$ is $\ker f \pmod{\ker f} = 1 \pmod{\ker f}$, so $\tilde{f}$ is injective. Also, $\text{im } \tilde{f} = \text{im } f$, so $\tilde{f}$ is onto $\text{im } f$. Therefore, the restriction $\tilde{f} : G / \ker f \rightarrow \text{im } f$ is bijective, and thus an isomorphism. ■

Corollary:

Let $f : G \rightarrow H$ be a homomorphism with G finite. Then $|\ker f| |\text{im } f| = |G|$.

Proof:

By the FIT, we know $|\text{im } f| = |G / \ker f| = |G| / |\ker f|$. ■

Conjugacy Classes and Centralizers

Every group G acts on itself by conjugation: $\phi : G \rightarrow \text{Sym}(G)$ by $\phi(g) = (x \mapsto gxg^{-1})$. The orbit of g under this action is called the conjugacy class of g . Note that $(x \mapsto gxg^{-1})$ is an isomorphism $G \rightarrow G$ (we already know it is a bijection, so we prove it's a homomorphism):

$$(\phi(g))(xy) = gxyg^{-1} = gxg^{-1}gyg^{-1} = \phi(g)(x) \cdot \phi(g)(y)$$

So elements of the same conjugacy classes are “algebraically identical”, for example they have the same order.

The stabilizer of x under this action is called the centralizer of x :

$$g \in \text{Stab}(x) \iff \phi(g)(x) = x \iff gxg^{-1} = x \iff gx = xg$$

so it's all the elements of G that commute with x .

Definition

The centre of a group G is the set

$$Z(G) = \{z \in G \mid zg = gz \forall g \in G\} = \bigcap_{g \in G} \text{Cent}(g)$$

In particular $Z(G)$ is a subgroup.

Example

Conjugacy in $GL_n(\mathbb{R})$ is similarity.

Example

If G is abelian, then conjugation is trivial, so conjugacy classes all have 1 element.

The Chinese Remainder Theorem and Identifying Products

Chinese Remainder Theorem

Define $\phi : \mathbb{Z}/mn\mathbb{Z} \rightarrow (\mathbb{Z}/m\mathbb{Z}) \times (\mathbb{Z}/n\mathbb{Z})$ by $\phi(x \pmod{mn}) = (x \pmod{m}, x \pmod{n})$, which is a

homomorphism. Both sides in the domain and codomain have mn elements, so if ϕ is injective we have an isomorphism. ϕ is injective if $\gcd(m, n) = 1$, because if $x \equiv 0 \pmod{m}$ and $x \equiv 0 \pmod{n}$ then $x \equiv 0 \pmod{mn}$. Thus, ϕ is injective under this hypothesis, and thus an isomorphism. So $\mathbb{Z}/mn\mathbb{Z} \cong (\mathbb{Z}/m\mathbb{Z}) \times (\mathbb{Z}/n\mathbb{Z})$ if $\gcd(m, n) = 1$.

When is $G \cong H \times K$? Well, subgroups of G would correspond to subgroups of $H \times K$. But $H' = \{1\} \times K$ and $K' = H \times \{1\}$ are both normal subgroups of $H \times K$ (kernels of maps like $(h, k) \mapsto h$), so $H' \cap K' = \{(1, 1)\}$. But $h'k' = k'h'$ for all $k' \in K'$, $h' \in H'$. Moreover, for every $x \in H \times K$ there are $h' \in H'$ and $k' \in K'$ satisfying $h'k' = x$. So if G is isomorphic to $H \times K$, then G has to have subgroups that are the images of H' and K' under this isomorphism that has to satisfy these properties.

Theorem

Let G be a group. Let M, N be normal subgroups of G satisfying:

(i) $M \cap N = \{1\}$.

(ii) $nm = mn$ for all $m \in M$ and $n \in N$.

(iii) For all $g \in G$, there exist $m \in M$ and $n \in N$ satisfying $mn = g$.

Then $G \cong M \times N$ via the map $\phi : M \times N \rightarrow G$ by $\phi(m, n) = mn$. The converse is true by our above discussion.

Proof:

Define $\phi : M \times N \rightarrow G$ by $\phi(m, n) = mn$. It's a homomorphism:

$$\phi((m_1, n_1)(m_2, n_2)) = \phi(m_1m_2, n_1n_2) = m_1m_2n_1n_2$$

But by property (ii), this is equal to

$$m_1n_1m_2n_2 = \phi(m_1, n_1)\phi(m_2, n_2)$$

ϕ is onto because of (iii). And it's injective because $\ker \phi = \{1\}$ because if $\phi(m, n) = 1$, then

$$mn = 1 \iff m = n^{-1}$$

but m and n^{-1} are in M and N respectively, and $M \cap N = \{1\}$, so we have $m = n = 1$. ■

Example

Say $G = D_n$, symmetries of a regular n -gon. Let $H \subseteq D_n$ be the subgroup of rotations. Is there a subgroup $N \subseteq D_n$ such that $D_n \cong H \times N$. No because H has n elements, so if such an N existed, it would have 2 elements. But H is abelian, and every group with 2 elements is isomorphic to $\mathbb{Z}_2$ and so is also abelian, but D_n isn't abelian. So it couldn't be isomorphic to the direct product of two abelian groups $H \times N$.

Hölder's Program:

If you understand N and G/N , then maybe you can understand G . Sadly, it's not that simple, with our above example: $D/H \cong \mathbb{Z}_2$, and $H \times \mathbb{Z}_2/(H \times \{1\}) \cong \mathbb{Z}_2$, but $D_n \not\cong H \times \mathbb{Z}_2$. Nevertheless, if a group G has a nontrivial normal subgroup N , then this idea has some merit: you can use N and G/N to understand G better.

Definition:

A group G is simple if its only normal subgroups are $\{1\}$ and G .

Example:

$\mathbb{Z}_p$ for p prime. If there were some normal subgroup, it would have to be a subgroup, which by Lagrange's Theorem must divide p , so by primality we have that this subgroup has order 1 or order p , i.e. is trivial or the whole group.

Free Groups, Generators & Relations

Definition

Let S be a set. The free group on S is the set of equivalence classes of finite strings

$$\{x_1 \dots x_r \mid r \in \mathbb{Z}_{\geq 0}, x_i = s \vee s^{-1}, s \in S\}$$

where the equivalence relation is the transitive closure of

$$x_1 \dots x_r \sim x_1 \dots x_m s s^{-1} x_{m+1} \dots x_r \sim x_1 \dots x_m s^{-1} s x_{m+1} \dots x_r$$

The operation is concatenation.

Examples:

If $S = \emptyset$, $F_S = \{1\}$. If $S = \{1\}$, then $F_S \cong \mathbb{Z}$, because F_S is strings of 1s and 1^{-1} 's (essentially strings of 1s and -1 s, which are just the integers). $F_{\{a,b\}}$ is huge. It has elements like $aba^{-1}b = a^2b^{-1}a^{-1}b^{-1}a^{-1}$, ab , ba and there's no other simplification you can do if a and a^{-1} (or b and b^{-1}) aren't next to each other.

Say $\langle g_1, g_2 \rangle \subseteq G$. Define $\phi : F_2 \rightarrow G$ by $\phi(\text{string})$ replacing all a 's with g_1 , all b 's with g_2 , and similarly for their inverses, then using G 's group operation. This is a homomorphism. Its image is $\text{im } \phi = \langle g_1, g_2 \rangle$, so if $G = \langle g_1, g_2 \rangle$ then it's onto. It's usually not injective, the kernel $\ker \phi$ is called the relations satisfied by g_1, g_2 .

Let F_S be the free group on S . Every group is the quotient of a free group: Say $G = \langle S \rangle$, where S is some subset of G . Then we can define a homomorphism $\phi : F_S \rightarrow G$ where ϕ maps a string to the same string, but as an element of G . This ϕ is onto because $\text{im } \phi$ contains S so it contains $\langle S \rangle = G$. So by the UPQ, ϕ induces an isomorphism $\tilde{\phi} : F_S / \ker \phi \rightarrow G$. We usually write this as $G = \langle S \mid R \rangle$ where R is a subset of F_S such that $\ker \phi$ is the smallest normal subgroup of F_S that contains R . This is called presenting G in terms of generators and relations.

Example:

$D_n = \{x, y \mid x^2, y^n, xyxy\}$, where we know every element of D_n is a reflection or rotation (x and y respectively), with $x^2 = 1$ (reflections applied twice are identity), $y^n = 1$ (rotations go all the way around), and $xyxy = 1$. Let's verify this:

Say $G = \{x, y \mid x^2, y^n, xyxy\}$. We want an isomorphism $G \xrightarrow{\cong} D$. There is a homomorphism $\phi : F_2 \rightarrow D_n$, where $\phi(x)$ is a reflection s , and $\phi(y)$ is a rotation r of $\frac{2\pi}{n}$ radians. This homomorphism contains x^2 , y^n and $xyxy$ in its kernel. So the UPQ gives a homomorphism. Letting N be the smallest normal subgroup generated by $\langle x^2, y^n, xyxy \rangle$, we get $\tilde{\phi} : F_2/N \rightarrow D_n$. $\tilde{\phi}$ is onto, because $\text{im } \tilde{\phi} = \text{im } \phi = D_n$. We want to show $\tilde{\phi}$ is injective. To do this, it's enough to show that F_2/N has at most $2n$ elements. An element of F_2/N is mN , where m is a string of x, y, x^{-1}, y^{-1} . We'll use the relations in R to do this. First, $x^2 = 1 \pmod{N}$ means that the string mN need not have consecutive x 's or x^{-1} 's, and similarly mN need not have strings of n or more y 's or y^{-1} . Using $xyxy = y^{-1}x$ ($x^{-1} = x$ since x is a reflection), we can see that mN can start with a string of x 's and end with a string of y 's. Thus, mod N , the string m can be written as y^i for $i = 0, \dots, n-1$, or xy^i , which are $2n$ possibilities. Therefore, since D_n has $2n$ elements and $\tilde{\phi}$ is surjective, this implies $\tilde{\phi}$ is injective as well, so we have an isomorphism $F_2/N \cong D_n$.

Symmetric Groups

S_n acts on $\mathbb{R}^n$ by permuting the coordinates. So we get a homomorphism $\phi : S_n \rightarrow \text{Sym}(\mathbb{R}^n)$. In fact, we get $\phi : S_n \rightarrow GL_n(\mathbb{R})$ (we can permute coordinates with a matrix). We can compose ϕ with $\det$ to get a homomorphism $\text{sgn}(\sigma) = \det(\phi(\sigma))$. Because a permutation matrix has determinant ± 1 , $\text{sgn}(\sigma) = \pm 1$. So $\text{sgn} : S_n \rightarrow \{\pm 1\}$ is a homomorphism whose kernel has $n!/2$ elements, since by Lagrange's Theorem

$$|\ker \text{sgn}| |\text{im sgn}| = |S_n| = n!.$$

Definition:

The subgroup $\ker \text{sgn} \subseteq S_n$ is called A_n , the alternating group on n letters. Elements of A_n are called even permutations, other elements of S_n are called odd permutations.

Example:

The identity (1) is even. Every transposition is odd. A 3-cycle $(xyz) = (xz)(xy)$ is a product of two odd transpositions, and thus is even. This leads to the general fact an n -cycle is odd if and only if n is even, and vice versa.

Proof:

$(a_1 \dots a_m) = (a_m a_{m-1}) \dots (a_2 a_1)$, a product of $n - 1$ odd permutations. ■

So the transpositions generate S_n , so any subgroup of S_n that contains all the 2-cycles must be S_n .

TO DO

Theorem (Orbit-Stabilizer):

Let G be a finite group acting on a set X . Let $x \in X$. Then $|\mathcal{O}_x| |\text{Stab}(x)| = |G|$.

Proof:

There is a homomorphism $\phi : G \rightarrow \text{Sym}(X)$. There is also a function $\psi : G \rightarrow \mathcal{O}_x$, given by $g \mapsto \phi(g)(x)$, where we have $\psi^{-1}(x) = \text{Stab}(x)$. For any $y \in \mathcal{O}_x$, there is $g \in G$ such that $\phi(g)(x) = y$. Therefore if $\phi(g')(x) = y$, then $\phi(g^{-1}g')(x) = \phi(g^{-1})(y) = x$, so $g^{-1}g' \in \text{Stab}(x)$. Conversely if $g^{-1}g' \in \text{Stab}(x)$, then $\phi(g^{-1}g')(x) = x$, implying $\phi(g^{-1})(\phi(g')(x)) = x$, implying $\phi(g')(x) = y$, so $\phi(g')(x) = y$ if and only if $g^{-1}g' \in \text{Stab}(x)$ if and only if $g' \in g\text{Stab}(x)$. Thus, for each element of $\mathcal{O}_x$, there are $|\text{Stab}(x)|$ elements of G satisfying $\phi(g)(x) = y$. Therefore $|G| = |\mathcal{O}_x| |\text{Stab}(x)|$. ■

Corollary

Let G act on itself by conjugation. If $x \in G$, then for this action, $\mathcal{O}_x$ is the conjugacy class of x and $\text{Stab}(x)$ is the centralizer of x , so $|\text{conj.class}| |\text{Cent}(x)| = |G|$.

Moreover, we have G to be the disjoint union of conjugacy classes. Say the conjugacy classes are $K_1, \dots, K_r$. Choose $g_i \in K_i$. Then

$$|G| = \sum_{i=1}^r |K_i| = \sum_{i=1}^r \frac{|G|}{|\text{Cent}(g_i)|} = |Z(G)| + \sum_{i=j+1}^r \frac{|G|}{|\text{Cent}(g_i)|}$$

where the equation $|G| = |Z(G)| + \sum_{i=j+1}^r \frac{|G|}{|\text{Cent}(g_i)|}$ is called the class equation. ■

Conjugacy classes of symmetric groups

Say $\tau \in S_n$. Then $\tau(a_1 \dots a_r) \tau^{-1} = (\tau(a_1) \tau(a_2) \dots \tau(a_r))$, so we see the conjugacy class of an r -cycle is the set of all r -cycles (since $(\tau(a_1) \dots \tau(a_r))$ is an arbitrary r -cycle). Applying this to disjoint cycle notation, we see conjugating σ by τ yields the same permutation only with a_i replaced by $\tau(a_i)$. So conjugacy classes in S_n are the sets of permutations with the same disjoint cycle “shape”.

Let's look at A_5 , the even permutations in S_5 .

These are:

$\{(1)\}$ - 1 element

$\{(abc)\}$ - $\binom{5}{3} \cdot 2 = 20$ ways

$\{(abcde)\}$ - $\binom{5}{5} \cdot 5!/5 = 24$ ways (5! ways we can order them, but we can cycle this order 5 ways)

$\{(ab)(cd)\}$ - $\binom{5}{2} \cdot \binom{3}{2}/2 = 15$

Adding these up, we get $60 = 5!/2$ cycles in A_5 .

We will use this list to show that A_5 is simple, i.e. its only normal subgroups are itself and the trivial subgroup. What are the conjugacy classes of A_5 ? (Any normal subgroup of A_5 is a union

of conjugacy classes). We have $|\text{conj.class}| |\text{Cent}| = |G|$. Using this formula, in S_5 , the conjugacy classes and centralizers are $\{1\} \leftrightarrow S_5$ and $\{(abc)\} \leftrightarrow$ a 6 element subgroup, $\{(abcde)\} \leftrightarrow \langle(abcde)\rangle$, $\{(ab)(cd)\} \leftrightarrow$ a 8 element subgroup.

But we know every conjugacy class in A_5 is contained in some conjugacy class of S_5 , and for all $\sigma \in A_5$, $\text{Cent}_{A_5}(\sigma) \subseteq \text{Cent}_{S_5}(\sigma)$. In fact, $\text{Cent}_{A_5}(\sigma) = \text{Cent}_{S_5}(\sigma) \cap A_5$. Therefore, we have $\text{sgn}(\text{Cent}_{S_5}(\sigma)) = \{\pm 1\}$ or $\{1\}$. If it's $\{1\}$, then $\text{Cent}_{S_5}(\sigma) = A_5$, so $\text{Cent}_{A_5}(\sigma)$. If it's $\{\pm 1\}$, then $\text{Cent}_{A_5}(\sigma)$ has index 2 in $\text{Cent}_{S_5}(\sigma)$ (by the First Isomorphism Theorem), so it's half the size.

Let's use this to look at our list of conjugacy classes: $\{(1)\}$'s centralizer is S_5 , so $\text{Cent}_{A_5}(\sigma)$ has size 60 so it's A_5 , so $\{(1)\}$ is a conjugacy class of A_5 .

$\{(abc)\}$ - this commutes with the odd 2-cycle (de) where $\{d, e\} \cap \{a, b, c\} = \emptyset$, so $\text{Cent}_{S_5}(abc) \not\subseteq A_5$ so $|\text{Cent}_{A_5}(abc)| = \frac{6}{2} = 3$, so the conjugacy class of (abc) has size $60/3 = 20$. Thus, the set $\{(abc)\}$ is a single conjugacy class in A_5 . The centralizer of $(abcde)$ is $\langle(abcde)\rangle \in A_5$, so the conjugacy class of $abcde$ has size $60/5 = 12$. So the S_5 conjugacy class of size 24 splits into 2 conjugacy classes of size 12 in A_5 .

REST OF PROOF OF A_5 SIMPLICITY (SEE LECTURE 15 PDF UPLOADED) ■

Definition

A p -group is a group whose order is a power of a prime p .

Definition

Let G be a finite group of order $p^a m$, where p does not divide m . A p -Sylow subgroup of G is a subgroup of order p^a .

Lemma:

Say $\phi : G \rightarrow H$ is a homomorphism. Then if $A \subseteq G$ is a subgroup, then $\phi(A)$ is a subgroup of H , and if $B \subseteq H$ is a subgroup, then $\phi^{-1}(B)$ is a subgroup of G .

Theorem (Sylow):

Let G be a group with $p^a m$ elements where p does not divide m .

- (1) G has a p -Sylow subgroup
- (2) Any p -subgroup of G is contained in a p -Sylow subgroup and any two p -Sylow subgroups are conjugate.
- (3) The number N of p -Sylow subgroups of G satisfies $N \equiv 1 \pmod{p}$, $N = [G : N(P)]$ divide $|G|$ where $N(P)$ is the normalizer of any p -Sylow subgroup.

Proof:

(1):

We induct on $|G|$. If G has 1 element, nothing to prove. If $|G| > 1$, there is some $p \mid |G|$ where p is a positive prime. If $p \mid |Z(G)|$, then there is some $x \in Z(G)$ with order p (see proof of a5 q5). Let $N = \langle g \rangle$. Then N has p elements and is normal because every subgroup of the center is normal. So G/N has fewer elements than G . So by the induction hypothesis, G/N has a p -Sylow subgroup $\bar{P}$. Let $P = \{g \in G : gN \in \bar{P}\}$. This is a subgroup of G by the lemma (it is the preimage of a subgroup in G/N under the reduction homomorphism), and since we know $\bar{P}$ has p^{a-1} elements, P has p^a elements, because P is the union of p^{a-1} cosets of size p . So P is a p -Sylow subgroup of G .

If $p \nmid |Z(G)|$, then the class equation says $|Z(G)| + \sum |\text{conj.class}| = |G|$. Then $p \mid |G|$, and there's some conjugacy class (say the class of x) whose size is not a multiple of p . But the sizes of the conjugacy classes must divide $|G|$ by Orbit-Stabilizer, so $|\text{Cent}(x)|$ must be divisible by p^a . But $\text{Cent}(x)$ is a proper subgroup of G because x is not in the center, so by induction $\text{Cent}(x)$ has a p -Sylow subgroup, which is also a p -Sylow subgroup of G (since it has the right size given our restriction on $\text{Cent}(x)$).

(2):

Say $P \subseteq G$ is a p -Sylow subgroup. Let $S = \{P_1, \dots, P_r\}$ be the conjugates of $P = P_1$. Let Q be a p -subgroup of G . Then Q acts on S by conjugation. Let $\mathcal{O}_1, \dots, \mathcal{O}_s$ be the orbits in S of the Q -action by conjugation. Reorder S so that $\mathcal{O}_i$ is the orbit of P_i (we might have $s < r$). By orbit-stabilizer,

$|\mathcal{O}_i| |\text{Stab}_Q(P_i)| = |Q|$. But $\text{Stab}_Q(P_i)$ is exactly $N_Q(P_i) = N_G(P_i) \cap Q$. Therefore, $|\mathcal{O}_i| = \frac{|Q|}{|N_G(P_i) \cap Q|}$. We will prove $N_G(P_i) \cap Q = P_i \cap Q$. The reverse inclusion is clear, since $P_i \subseteq N_G(P_i)$. Let's prove $N_G(P_i) \cap Q \subseteq P_i \cap Q$. Write $H = N_G(P_i) \cap Q$. It's enough to show that $HP_i \subseteq P_i$. First, notice that HP_i is a group. First, notice that $1 \in HP_i$. Next, $h_1 p_1, h_2 p_2 \in HP_i$ implies $h_1 p_1 h_2 p_2 = h_1 h_2 p'_i p_2 \in HP_i$ for some $p'_i \in P_i$ because $h_2^{-1} p_1 h_2 \in P_i$ because $H \subseteq N_G(P_i)$, and $(hp)^{-1} = p^{-1} h^{-1} = h^{-1} p'$ for some $p' \in P_i$ by similar reasoning.

The number of elements of H and P_i are both powers of p , so $|HP_i|$ must also be a power of p . Therefore, HP_i is a p -subgroup that contains P_i , but P_i is a maximal p -subgroup (since it is Sylow), so $HP_i \subseteq P_i$. Therefore, $|\mathcal{O}_i| = \frac{|Q|}{|P_i \cap Q|}$. But this works for any p -subgroup Q , like say, $P = P_1$. Then $|\mathcal{O}_1| = \frac{|P_1|}{|P_1|} = 1$. If $i \neq 1$, then $|\mathcal{O}_i| = \frac{|P_1|}{|P_1 \cap P_i|} > 1$ (since $P_1 \cap P_i \subsetneq P_1$), and since $|P_1|$ is a power of p , this tells us $p \mid |\mathcal{O}_i|$.

Therefore, $|\mathcal{O}_1| + |\mathcal{O}_2| + \dots + |\mathcal{O}_s| \equiv 1 \pmod{p}$, i.e. the number of conjugates of P is congruent to 1 mod p . For any old Q , if $Q \not\subseteq P_i$, then p divides $\frac{|Q|}{|Q \cap P_i|} = |\mathcal{O}_i|$ (since $Q \cap P_i \subsetneq Q$). Therefore, we must have $Q \subseteq P_i$ for some i , because $p \nmid (|\mathcal{O}_1| + \dots + |\mathcal{O}_s|)$, and if it were not for any i , then we would have $p \mid |\mathcal{O}_1| + \dots + |\mathcal{O}_s|$ which is impossible by our congruence above (this is the first part of (2)).

So if P' is any p -Sylow subgroup of G , then we can set $Q = P'$ to get $P' \subseteq P_i$ for some i , so by maximality of P' , we get $P' = P_i$, so all p -Sylow subgroups are conjugate. This proves (2). Moreover, our above congruence shows the number of conjugates of P (the number of p -Sylow subgroups) is congruent to 1 mod p , which is the first part of (3).

Now, by orbit-stabilizer, the number of p -Sylows multiplied by $N_G(P)$ is equal to $|G|$, so the number of p -Sylows is equal to $[G : N_G(P)]$. ■

Definition

Let $S \subseteq G$. The normalizer of S in G is $N_G(S) = N(S) = \{g \in G : gsg^{-1} \in S \forall s \in S\}$.

In particular, if S is a subgroup, S is normal in $N(S)$, and any subgroup H of which S is a normal subgroup satisfies $H \subseteq N_G(S)$.

Example

Let G be a group of order 15. Let H be a 3-Sylow subgroup. Then $|H| = 3$. Let K be a 5-Sylow subgroup. Then $|K| = 5$. The number of conjugates of H is congruent to 1 mod 3 and divides 15, so H has one conjugate, so H is normal. Similarly, K is normal. Since $\gcd(|H|, |K|) = 1$, we have $H \cap K = \{1\}$ (by Lagrange's Theorem, any element in H would have to have order dividing 3, similarly 5 for K , so any element in both has order 1).

Consider $HK = \{hk : h \in H, k \in K\}$. If $h_1 k_1 = h_2 k_2$, then $h_2^{-1} k_1 = k_2^{-1} k_1 \in H \implies h_2 h_1^{-1} = 1 \implies h_1 = h_2$ and $k_1 = k_2$. Therefore, we must have $HK = G$, since all 15 hk are distinct, and there are 15 elements in G . Moreover, $hk = kh$ for all $h \in H$ because $hkh^{-1}k^{-1} \in H \cap K$ because H and K are normal so this product has an element of H times a conjugate of an element of H (similarly for K). Put together, we meet all the hypotheses to say $G \cong H \times K$.

Classification of finite groups of small order

$|G| = 1$ implies $G = \{1\}$.

$|G| = 2$ implies $G = \langle x \rangle$ for $x \in G$, $x \neq 1$, x of order 2, so $G \cong \mathbb{Z}_2$.

Similarly, $|G| = 3$ implies $G \cong \mathbb{Z}_3$.

$|G| = 4$ implies G is abelian by a homework assignment since $4 = 2^2$. If G has an element x of order 4, $G = \langle x \rangle$, so $G \cong \mathbb{Z}_4$, but if it does not have an element of order 4, then G has three elements of order 2, let x, y be two of them. Then $G = \langle x, y \rangle = \langle x \rangle \langle y \rangle$ (since this is a subgroup of G with at least 3 elements, and thus must have 4 elements since 3 does not divide 4) and $\langle x \rangle \cap \langle y \rangle = \{1\}$, so $G \cong \langle x \rangle \times \langle y \rangle \cong \mathbb{Z}_2 \times \mathbb{Z}_2$.

$|G| = 5$ implies $G \cong \mathbb{Z}_5$.

If $|G| = 6$, there is a 2-Sylow P_2 and 3-Sylow P_3 . The subgroup P_3 is normal because 1 is the only positive integer congruent to 1 mod 3 that divides 6, so there can be only 1 3-Sylow subgroup and

so P_3 has no conjugates, so it is normal. If P_2 is also normal, then $G \cong P_2 \times P_3 \cong \mathbb{Z}_2 \times \mathbb{Z}_3 \cong \mathbb{Z}_6$ by the Chinese remainder theorem. If it is not, then there are 3 conjugates of P_2 , call them A, B, C . Then G acts on $\{A, B, C\}$ transitively by conjugation. So there is a homomorphism $\phi : G \rightarrow S_3$. The kernel of ϕ is the set $\{g \in G : gAg^{-1} = A, gBg^{-1} = B, gCg^{-1} = C\}$. The G -orbit has size 3, so each stabilizer has size 2. Since $A \subseteq \text{Stab}(A)$, so $A = \text{Stab}(A)$ and A is size 2, we get $A = \text{Stab}(A)$, so $\text{Stab}(A) \cap \text{Stab}(B) = A \cap B = \{1\}$. Thus the kernel of ϕ is $\{1\}$, so ϕ is an isomorphism (it is a bijection since it is between two sets of size 6, and we just proved it is injective), so $G \cong S_3$.

If $|G| = 7$, $G \cong \mathbb{Z}_7$.

If $|G| = 8$, and if G has an element x of order 8, then $G = \langle x \rangle$ and $G \cong \mathbb{Z}_8$. If every nontrivial element has order 2, then $G \cong \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$, as it is abelian, since $(ab)^2 = 1 \implies abab = 1 \implies ab = ba$ (since elements are their own inverses by order 2), so three elements a, b, c with $a \neq b \neq c$, $abc \neq 1$ generate subgroups satisfying $G \cong \langle a \rangle \times \langle b \rangle \times \langle c \rangle$. Now say $x \in G$ has order 4. Then $\langle x \rangle$ has index 2, and so it's normal: Let $H \subseteq G$ be a subgroup of index 2. Then H is normal: Define $\phi : G \rightarrow \{\pm 1\}$ by $\phi(g) = 1$ if $g \in H$, -1 otherwise. This is a homomorphism since if $g_1, g_2 \notin H$, then $g_1g_2 \in H$ because the stabilizer of H under the G -action by left multiplication is H , and similarly for $G \setminus H$, so $g_1g_2H = g_1(g_2H) = g_1(G \setminus H) = H$ (since index 2 means that to switch cosets, we switch between H and $G \setminus H$), so $g_1g_2 \in \text{Stab}(H) = H$, so ϕ is a homomorphism so $H = \ker \phi$ is normal. Say $y \notin \langle x \rangle$. Then $xyx^{-1} = x^a$, so x^a has order 4 (since x does). That means $a = 1$ or -1 . If $a = 1$, $yx = xy$, since $G = \langle x, y \rangle$, that means G is abelian, and a bit of work (or a massive theorem we'll prove in a few weeks) shows $G \cong \mathbb{Z}_4 \times \mathbb{Z}_2$. Thus, say $a = -1$, so $yx = x^{-1}y$. If y has order 2, then $G = \{x, y : x^4 = y^2 = 1, yx = x^{-1}y\} \cong D_4$. The only other possibility is that every element of $G \setminus \{1, x\}$ has order 4. So $G = \{1, -1, x, x^{-1}, y, y^{-1}, z, z^{-1}\}$, where 1 has order 1, -1 has order 2, the rest have order 4. Since -1 is the only element of order 2, we must have $-1 = x^2 = y^2 = z^2$, so every element commutes with -1 . With this rule, this is clearly isomorphic to the quaternions (as can be seen with a multiplication table), so $G \cong Q_8$.

For $|G| = 9 = 3^2$, so G is abelian. By the same argument for 4, it must be $\mathbb{Z}_9$ or $\mathbb{Z}_3 \times \mathbb{Z}_3$.

For $|G| = 10$, this is the same as 6, where $G \cong \mathbb{Z}_{10}$ or $G \cong D_5$ (depending on whether the 2-Sylow subgroup is normal).

For $|G| = 11$, $G \cong \mathbb{Z}_{11}$.

Now consider the case $|G| = 12 = 2^2 \cdot 3$. Let P_2 be a 2-Sylow, and let P_3 be a 3-Sylow. If P_2, P_3 are both normal, then $G \cong P_2 \times P_3$, so $G \cong \mathbb{Z}_{12}$ or $G \cong \mathbb{Z}_2 \times \mathbb{Z}_6$. So let's say one of P_2 or P_3 are not normal. If P_3 is not normal, then there are 4 3-Sylow subgroups, and so G acts transitively on the set of four conjugates $\{A, B, C, D\}$ of P_3 by conjugation. So there exists a homomorphism $\phi : G \rightarrow S_4$. The stabilizer of P_3 is just P_3 , by the orbit stabilizer theorem, since the size of its orbit is 4 (the other conjugates, because the action is transitive), so the size of its stabilizer must be $12/4 = 3$. Therefore, the kernel of this homomorphism is the intersections of the stabilizers of A, B, C, D under this action, and the same argument applies for those, and those have trivial intersection, so $\ker \phi = \{1\}$ so ϕ is injective. Therefore, G must be isomorphic to an order 12 subgroup of S_4 , so G must be isomorphic to A_4 , since A_4 is the only 12-element subgroup of S_4 (it is the only normal 12-element subgroup of S_4 , and any 12-element subgroup must be normal by index 2), so $G \cong A_4$. The only case left is that P_3 is normal but P_2 isn't. Because $|P_2| = 4$, either $P_2 \cong \mathbb{Z}_4$ or $P_2 \cong \mathbb{Z}_2 \times \mathbb{Z}_2$. Suppose $P_2 \cong \mathbb{Z}_4$, so $P_2 = \langle y \rangle$ and $P_3 = \langle x \rangle$ for $x, y \in P_2$. Since P_3 is normal, $xyx^{-1} = x^a$ for some a . Either $a = 1$ or $a = -1$, if $a = 1$, then $yx = xy$ and G is abelian, which can't be the case since it has a non-normal subgroup. So we may assume $a = -1$, so $yx = x^{-1}y$. So $G \cong \{x, y \mid x^3 = y^4 = 1, yx = x^{-1}y\}$. To see this, write $\phi : F_2 \rightarrow G$, $\phi(x) = x$, $\phi(y) = y$. If $N \subseteq F_2$ is the normal subgroup generated by $\{x^3, y^4, yxy^{-1}x\}$, then $N \subseteq \ker \phi$, so ϕ induces $\tilde{\phi} : F_2/N \rightarrow G$ by the UPQ. $\tilde{\phi}$ is surjective because $G = \langle x, y \rangle$, we want to show it's injective. It's one-to-one because F_2/N has at most 12 elements, using $x^3 = y^4 = 1$ and $yx = x^{-1}y$, we can represent any element of F_2/N by $x^a y^b$ for $a \in \{0, 1, 2\}$ and $b \in \{0, 1, 2, 3\}$. Finally, say P_3 is normal and $P_2 \cong \mathbb{Z}_2 \times \mathbb{Z}_2$. Write $P_2 = \{1, a, b, c\}$, where a, b, c all have order 2 and $c = ab$, and write $P_3 = \langle x \rangle$. So $axa^{-1} = axa = x^k$ for some $k \in \{-1, 1\}$. If $k = 1$, then the group $\langle a, x \rangle$ is abelian with 6 elements, so it's isomorphic to $\mathbb{Z}_6$, so $\langle a, x \rangle = \langle z \rangle$ for some $z \in \langle a, x \rangle$. Now, $bzb^{-1} = z^\ell$, where $\ell \in \{-1, 1\}$ (because z has order 6). If $\ell = 1$, then G is abelian, and we reduce to the previous case. If $\ell = -1$, then $G \cong D_6$, because $G \cong \{b, z \mid b^2 = z^6 = 1, bz = z^{-1}b\}$. This works if we got k we

conjugating x by any of a, b, c , provided at least one gives $k = 1$, and $k = -1$ isn't actually possible for all 3. If $k = -1$, consider $bzb^{-1} = z^\ell$. If $\ell = 1$, then the previous argument shows $G \cong D_6$. If $\ell = -1$, then $bzb^{-1} = z^{-1}$. So $czc^{-1} = abzb^{-1}a^{-1} = az^{-1}a^{-1} = z$, so by the previous argument, $G \cong D_6$.

If $|G| = 13$, $G \cong \mathbb{Z}_{13}$.

If $|G| = 14$, $G \cong \mathbb{Z}_{14}$ or $G \cong D_7$ by the argument for 10 or 6.

If $|G| = 15$, a previous example shows $G \cong \mathbb{Z}_{15}$. ■

Let's try to find a group presentation for S_4 : $S_4 = \langle x, y \rangle$, with $x = (12)$ and $y = (234)$. The brute force way of doing this is to write down strings of x and y to find their relations. We already know $x^2 = 1$, $y^3 = 1$ (we can use these to not take any inverses, since $x = x^{-1}$, $y^2 = y^{-1}$). The strings of x and y of length 0 are (1) , the strings of length 1 are $(12) = x = x^{-1}$, $(234) = y$. Multiplying these on the left by x and y , we additionally get $xy = (1234)$, $yx = (1342)$, $y^2 = y^{-1} = (243)$. Multiplying these on the left by x and y , $x^2y = y$ is a consequence of a relation we already had, $xyx = (134)$, $xy^2 = (1243)$, $xyxy = (1324)$, $y^2x = (1432)$. Multiplying these on the left by x and y , $xyxy = (13)(24)$, $xy^2x = (143)$, $xyxyx = (1423)$, $xyxy^2 = (13)$, $y^2xy = (14)$. To get the strings of length 5, we multiply on the left again, $xyxyx = (1423)$, $xyxy^2 = (132)$, $xy^2xy = (142)$, $y^2xyx = (124)$, $xyxy^2x = (123)$, $xyxyxy = (143)$, which repeats as xy^2x , so we see $xyxyxy = xy^2x$ which tells us $xyxyxyx = y^2$, which tells us $(xy)^4 = 1$, which tells us $(yx)^4 = 1$. There are 4 more elements, and in general, we look at all elements we've generated (by looking at all the possible finite strings until we have as many distinct elements as the order of our group) and any relations that have shown up, and try to argue that these relations are sufficient by now counting the number of elements of the given group presentation. For S_4 , these end up being enough. ■

Rings

A ring is a bunch of things you can add, subtract, and multiply.

Examples

The integers $\mathbb{Z}$, the rationals $\mathbb{Q}$, the reals $\mathbb{R}$, the complex numbers $\mathbb{C}$, the real polynomials in any number of variables like $\mathbb{R}[x]$, $\mathbb{R}[x, y]$, $\mathbb{R}[x_1, \dots, x_n]$ (and nothing is special about $\mathbb{R}$ here, we can take any ring). The integers modulo n , $\mathbb{Z}/n\mathbb{Z}$. The $n \times n$ matrices $M_n(\mathbb{R})$ (where we can take any ring as the entries). This last ring is a little different from the past ones, because its multiplication isn't commutative, motivating the definition that a ring is commutative if and only if $ab = ba$ for all a, b .

Definition

A ring is a set R with two operations $+: R \times R \rightarrow R$ and $\cdot: R \times R \rightarrow R$ satisfying (for all $a, b, c \in R$):

- (i) $(a + b) + c = a + (b + c)$
- (ii) $a + b = b + a$
- (iii) There exists $0 \in R$ such that $0 + a = a$
- (iv) There is a $-a \in R$ such that $a + (-a) = 0$
- (v) $(a \cdot b) \cdot c = a \cdot (b \cdot c)$
- (vi, this is nonstandard) There exists $1 \neq 0 \in R$ such that $a \cdot 1 = 1 \cdot a = a$ for all $a \in R$.
- (vii) $a \cdot (b + c) = a \cdot b + a \cdot c$ and $(a + b) \cdot c = a \cdot c + b \cdot c$.

Definition

A ring R is commutative if and only if $ab = ba$ for all $a, b \in R$.

Definition

A ring R is a division ring if and only if for all $a \in R$, $a \neq 0$, there is an $a^{-1} \in R$ such that $aa^{-1} = a^{-1}a = 1$.

Definition

A field is a commutative division ring.

Definition

An element $a \in R$ is called a unit if and only if there exists $a^{-1} \in R$ such that $aa^{-1} = a^{-1}a = 1$.

Definition

An element $a \in R$ is called a zero divisor if and only if $a \neq 0$ and there is some $b \neq 0 \in R$ such that $ab = 0$ or $ba = 0$.

Definition

An integral domain, or domain, is a ring with no zero divisors.

Example

$\mathbb{Z}/6\mathbb{Z}$ is not a domain: $2 \neq 0, 3 \neq 0, (2)(3) = 6 = 0$, so 2 and 3 are zero divisors.

Theorem

Every unit is not a zero divisor.

Proof:

Suppose $a \in R$ is a unit, and say $ab = 0$. Then $b = a^{-1} \cdot 0 = 0$, and $ba = 0 \implies b = 0$, so a is not a zero divisor. ■

Corollary:

Every division ring is an integral domain. ■

Example

In $\mathbb{Z}$, the units are $\{\pm 1\}$. $\mathbb{Q}, \mathbb{R}, \mathbb{C}$ are fields. $M_n(\mathbb{R})$ is not commutative, and has zero divisors if $n \geq 2$, so it's not a domain and thus not a division ring. Its units are $GL_n(\mathbb{R})$. $\mathbb{R}[x]$ is commutative and a domain, but not a field, as its units are only the nonzero constants.

Definition

A subring of a ring R is a subset $S \subseteq R$ that is a ring under the same operations and 1 of R .

Theorem (Subring):

A subset $S \subseteq R$ where R is a ring is a subring if and only if

- (i) $1 \in S$
- (ii) S is closed under $-$, that is $a, b \in S$ implies $a - b \in S$
- (iii) S is closed under $\cdot$, that is $a, b \in S$ implies $ab \in S$.

Proof:

The reverse direction is clear, since if S is a subring, all three statements hold trivially. So assume S satisfies (i), (ii), (iii). First, note that $\cdot : S \times S \rightarrow S$ is well-defined by (iii). Since $1 \in S$ by (i), by (ii), we get $0 = 1 - 1 \in S$, so $-1 = 0 - 1 \in S$. So, if $a, b \in S$, then $-b \in S$ by (ii), so $a + b = a - (-b) \in S$, so we see $+$ is well-defined. Associativity of $+$ and $\cdot$ and commutativity of $+$, as well as the two distributive axioms, are immediately true for S , since they are true for their superset R . Existence of 0, additive inverses, and 1 in S follow from our above justification. ■

Example

Let $R = \mathbb{C}$, let S be $S = \{a + b\gamma + c\gamma^2 + d\gamma^3 + e\gamma^4\}$, where $a, b, c, d, e \in \mathbb{Z}$, $\gamma = e^{2\pi/5}$ (which satisfies $\gamma^5 = 1$), which we usually write as $S = \mathbb{Z}[\zeta_5]$. Using the Subring theorem, we see (i) holds since we can take $a = 1, b = c = d = e = 0$. (ii) holds because of the distributive property of $\mathbb{Z}$. For (iii), say $x \in S, y \in S$, $x = a + b\gamma + c\gamma^2 + d\gamma^3 + e\gamma^4$, $y = a' + b'\gamma + c'\gamma^2 + d'\gamma^3 + e'\gamma^4$, TODO

TODO DEFINITIONS OF HOMOMORPHISM, ISOMORPHISM, IMAGE, KERNEL, HOMOMORPHISMS PRESERVE 1

Theorem

The image of a homomorphism $\phi : R \rightarrow S$ is a subring of S , the kernel is not a subring.

Proof:

the first statement is clear, the second statement happens because the kernel doesn't contain 1 unless it's a trivial map, but we see in the proof that the kernel IS "acted upon by R -multiplication".

Definition

A (left) R -module M is an abelian group $(M, +)$ with an action $\cdot : R \times M \rightarrow M$ that is compatible with $+$ with distributivity (a bunch of things you can add and subtract, and multiply by elements of R).

From now on, every ring we deal with is commutative.

Example

If R is a field F , then an R -module is exactly an F -vector space.

Example

$2\mathbb{Z}$ (the even integers) is a $\mathbb{Z}$ -module.

Example

$\mathbb{Z}/6\mathbb{Z}$ is a $\mathbb{Z}$ -module.

Definition

A submodule of an R -module M is a subset $S \subseteq M$ that is an R -module using the same $+, -, \cdot$ as M .

Theorem (Submodule):

A subset S of an R -module M is an R -submodule of M if and only if

- (i) $0 \in S$.
- (ii) S is closed under subtraction.
- (iii) S is closed under $\cdot$.

Definition

An ideal of a ring R is an R -submodule of R .

Example

$2\mathbb{Z}$ is an ideal of $\mathbb{Z}$.

We showed, last time that if $\phi : R \rightarrow T$ is a homomorphism, then $\ker \phi$ is an ideal of R . Is it true that every ideal of R is the kernel of some homomorphism? Yes, we'll take the quotient. Let's say $I \subseteq R$ is an ideal. We want to find a homomorphism $\phi : R \rightarrow T$ with $\ker \phi = I$. If we had such a ϕ and T , then $\phi^{-1}(0) = I$, $\phi^{-1}(1) = 1 + I$, and in general $\phi^{-1}(t) = r + I$ where $\phi(r) = t$. These look awfully familiar. So define R/I to be $\{r + I \mid r \in R\}$, with multiplication $(r_1 + I) + (r_2 + I) = (r_1 + r_2) + I$. $(r_1 + I)(r_2 + I) = r_1 r_2 + I$, and $1 + I$ is the multiplicative identity, $0 + I$ is the additive identity. It is proven in the textbook that R/I is a ring. R/I is NOT a subring.

Theorem: (Universal Property of Quotients)

Let $\phi : R \rightarrow T$ be a homomorphism, $I \subseteq R$ an ideal. Then there exists a homomorphism $\tilde{\phi} : R/I \rightarrow T$ satisfying $\phi = \tilde{\phi} \circ \pi$ (where π is the reduction modulo I homomorphism) if and only if $I \subseteq \ker \phi$. Furthermore, $\text{im } \tilde{\phi} = \text{im } \phi$, and $\ker \tilde{\phi} = \pi(\ker \phi)$. The proof is analogous to the proof for groups.

Example

Take $R = \mathbb{R}[x]$. $I = \{p(x) \mid p(1) = 0\}$ is an ideal. What does $\mathbb{R}[x]/I$ look like? Define $\phi : \mathbb{R}[x] \rightarrow \mathbb{R}$ by $\phi(p(x)) = p(1)$. As proved previously, this is a homomorphism (since it is an evaluation homomorphism). It is easy to see I is exactly $\ker \phi$. Therefore, by the UPQ, $\tilde{\phi} : \mathbb{R}[x]/I \rightarrow \mathbb{R}$ has image $\mathbb{R}$ and kernel $\{0 + I\}$, so it's injective and surjective (the proof of a kernel being trivial implies injectivity is the same as for groups or vector spaces), so it's an isomorphism.

Theorem

A ring homomorphism $\phi : R \rightarrow T$ is injective if and only if $\ker \phi = \{0\}$.

Proof:

The forward direction is clear.

In the reverse direction, suppose $\ker \phi = \{0\}$, and suppose $x \neq y \in R$. Then $\phi(a - b) = 0$, so $a - b \in \ker \phi$ implying $a = b$. ■

Definition

An ideal $I \subseteq R$ is maximal if and only if $I \neq R$ and if $J \subseteq R$ is an ideal with $I \subseteq J \subseteq R$, then either $J = I$ or $J = R$.

Example

Let $R = \mathbb{Z}$. What are the ideals of R ? Say $I \subseteq \mathbb{Z}$ is an ideal. We could have $I = \{0\} = (0)$. If $I \neq (0)$, then there is some $n \in I$, $n \neq 0$. Let's choose the smallest positive $n \in I$. Then we claim $I = n\mathbb{Z}$. We have $n\mathbb{Z} \subseteq I$ because $n \in I$ and closure of I under multiplication by $\mathbb{Z}$. So we just need to show $I \subseteq n\mathbb{Z}$. Let $x \in I$. Write $x = qn + r$ where $q, r \in \mathbb{Z}$ and $0 \leq r < n$. Then $r = x - qn \in I$. But now, by minimality of n , this tells us $r = 0$. So $x = qn \in n\mathbb{Z}$, completing the proof of this claim. So every ideal of $\mathbb{Z}$ is of the form $n\mathbb{Z}$ for some $n \in \mathbb{Z}$. And $n\mathbb{Z} \subseteq k\mathbb{Z}$ if and only if $k \mid n$. So $n\mathbb{Z} \subseteq \mathbb{Z}$ is maximal if and only if n is prime.

Notation:

Let R be a ring, $S \subseteq R$ any subset. The ideal generated by S , denoted (S) , is the intersection of all ideals that contain S . More concretely, $(S) = \{r_1 s_1 + \cdots + r_n s_n \mid r_i \in R, s_i \in S\}$. When $S = \{x\}$, then $(x) = \{rx \mid r \in R\}$.

Example

We have $(1) = R$ for every ring.

Example

For $(6, 8) \subseteq \mathbb{Z}$, $(6, 8) = \{6a + 8b \mid a, b \in \mathbb{Z}\}$. We know this is (n) for some $n \in \mathbb{Z}$ by our example above. n is even since $6a + 8b$ is always even, and we know $2 \in (6, 8)$, so we have $(2) \subseteq (6, 8)$ and $(6, 8) \subseteq (2)$, so $(6, 8) = (2) = 2\mathbb{Z}$.

Theorem

An ideal $I \subseteq R$ is maximal if and only if R/I is field.

Lemma:

The ideals of R/I are precisely the reductions mod I of ideals of R that contain I .

Proof:

Say $J \subseteq R$ is an ideal that contains I . Then if $\pi : R \rightarrow R/I$ is the quotient homomorphism, $\pi(J)$ is an ideal of R/I because the image of an ideal under a homomorphism is an ideal.

Conversely, if $\bar{J}$ is an ideal of R/I , define $J = \{r \in R \mid \pi(r) \in \bar{J}\} = \pi^{-1}(\bar{J})$. This is an ideal: $0 \in J$ since $\pi(0) = 0 \in \bar{J}$, if $x, y \in J$, then $\pi(x - y) = \pi(x) - \pi(y) \in \bar{J}$ so $x - y \in J$. If $r \in R$ and $x \in J$, then we want to verify $rx \in J$. But $\pi(rx) = \pi(r)\pi(x) \in \bar{J}$ since $\pi(r) \in R/I$ and $\pi(x) \in \bar{J}$, so $rx \in J$. Finally, note that if $x \in I$, then $\pi(x) = 0 \in \bar{J}$, so $x \in J$, so $I \subseteq J$. ■

As a consequence of this lemma, we see if $\bar{J}_1 \neq \bar{J}_2$ then $J_1 \neq J_2$ because π is onto.

Proof of theorem about maximal ideals:

In the reverse direction, suppose R/I is a field. We want to show I is maximal. First note that for any ideal, if it contains a unit, then that ideal is the whole ring, and thus contains 1 by closure (multiply by a^{-1}), so by the above example it contains the whole ring. Any nonzero ideal of R/I contains a unit since R/I is a field, so any nonzero ideal of R/I is R/I . Thus, the only two ideals of R/I are zero and R/I . So by the lemma, the only ideals of R that contain I are I and R , so I is maximal.

In the forward direction, suppose I is a maximal ideal. By the lemma, R/I has exactly two ideals, (0) and R/I . Let $x \in R/I$ be any nonzero element. Then $(x) = R/I$, in particular, $1 \in (x)$ so $1 = rx$ for some $r \in R/I$, so x is a unit, so every nonzero element of R/I is a unit and R/I is a field. ■

Corollary:

The maximal ideals of $\mathbb{Z}$ are the ideals $(p) = p\mathbb{Z}$ for p prime by our above example. So by the theorem, $\mathbb{Z}/n\mathbb{Z}$ is a field if and only if n is prime. ■

Example

Say F is a field. What are the maximal ideals of $F[x]$? First, say $I \subseteq F[x]$ is an ideal. We could have $I = (0)$. If not, then there is some $p(x) \in I$ for $p(x) \neq 0$. Let $p(x)$ be a nonzero polynomial of minimal degree in I . We'll show $(p(x)) = I$, which will be analogous to our proof for $\mathbb{Z}$ above. Say $q(x) \in I$. We want to show $q(x) = t(x)p(x)$ for some $t(x) \in F[x]$. By the division algorithm, take $q(x) = t(x)p(x) + r(x)$ for $r(x)$ having degree less than $p(x)$'s. But $r(x) \in I$ by closure of I and $p(x) \in I$, so $r(x) = 0$ by minimality of $p(x)$'s degree, so $r(x) = 0$ and $q(x) = t(x)p(x)$, so $q(x) \in (p(x))$, so $I = (p(x))$. ■

Theorem

Let $\phi : F \rightarrow T$ be a homomorphism where F is a field. Then ϕ is injective.

Proof:

$\ker \phi$ is an ideal of F . So $\ker \phi = (0)$ or (1) . But $\phi(1) = 1$, so $1 \notin \ker \phi$, so we must have $\ker \phi = (0)$ so ϕ is injective. ■

Now, we have seen quotients by maximal ideals correspond to fields. Do any ideals correspond to domains?

Let a, b be elements of a domain. Then $ab = 0$ implies $a = 0$ or $b = 0$. So R/I is a domain if and only if $ab \equiv 0 \pmod I$ implying $a \equiv 0 \pmod I$ or $b \equiv 0 \pmod I$ if and only if $ab \in I$ implies $a \in I$ or $b \in I$.

Definition

An ideal $I \subsetneq R$ is prime if and only if for all $a, b \in R$ with $ab \in I$, either $a \in I$ or $b \in I$.

Theorem

R/I is a domain if and only if I is a prime ideal. The proof is our above calculation. ■

Since every field is a domain, every maximal ideal is prime.

Example

What are the prime ideals of $\mathbb{Z}$? We showed $n\mathbb{Z}$ is maximal if and only if n is prime. We see $n\mathbb{Z}$ is prime if and only if n is prime or $n = 0$.

Definition

A principal ideal domain is a domain D such that every ideal of D can be generated by exactly one element.

Example

$\mathbb{Z}$ is a PID since we showed every ideal is of the form (n) . Similarly, we showed every ideal of $F[x]$ where F is a field is a PID (a polynomial of minimal degree in the field).

Let R be any ring. There is a unique homomorphism $\phi : \mathbb{Z} \rightarrow R$ called the characteristic homomorphism, defined by $\phi(n) = n1$. The kernel of ϕ is $n\mathbb{Z}$ for some $n \in \mathbb{Z}$ since the kernel is an ideal. We may assume $n \geq 0$, because $n\mathbb{Z} = -n\mathbb{Z}$. The value of n is called the characteristic of R .

Example

If $R = \mathbb{Z}$, then $\text{char } \mathbb{Z} = 0$, because the kernel of the identity homomorphism is 0. If $R = \mathbb{Q}$, we have $\text{char } \mathbb{Q} = 0$. If $R = \mathbb{Z}/n\mathbb{Z}$, we have $\text{char } \mathbb{Z}/n\mathbb{Z} = n$.

Fact:

If D is a domain, then $\text{im } \phi$ is also a domain, so by the first isomorphism theorem, $\ker \phi$ is a prime ideal of $\mathbb{Z}$, so $\text{char } D = 0$ or a prime number.

Definition

Let's say R is a ring, T is a ring that contains R . Let $\alpha \in T$. Then $R[\alpha] = \{a_n \alpha^n + \cdots + a_1 \alpha + a_0 \mid n \in \mathbb{N}, a_i \in R\}$.

Example

Let $\zeta_5 = e^{2\pi i/5} \in \mathbb{C}$. Then $\mathbb{Z}[\zeta_5] = \{a_n \zeta_5^n + \cdots + a_0 \mid a_i \in \mathbb{Z}\} = \{a_0 + a_1 \zeta_5 + a_2 \zeta_5^2 + a_3 \zeta_5^3 + a_4 \zeta_5^4 \mid a_i \in \mathbb{Z}\}$.

Example

$\mathbb{Z}[i] = \{a_n i^n + \cdots + a_0 \mid a_i \in \mathbb{Z}\} = \{a_1 i + a_0 \mid a_i \in \mathbb{Z}\}$.

Example

$\mathbb{Z}[\sqrt{2}, \sqrt{3}] = \{p(\sqrt{2}, \sqrt{3}) \mid p(x, y) \in \mathbb{Z}[x, y]\} = \{a_0 + a_{10}\sqrt{2} + a_{01}\sqrt{3} + a_{11}\sqrt{6} \mid a_{ij} \in \mathbb{Z}\}$.

Let F be a field. What are the ideals of $F[x]$? They are all of the form $(p(x))$. The ideal $(p(x))$ is

maximal when there are no ideals T with $(p(x)) \subsetneq T \subsetneq F[x]$, but $(p(x)) \subseteq (q(x))$ if and only if $q(x) \mid p(x)$, so $p(x)$ is maximal if and only if $p(x)$ has no nontrivial factors in $F[x]$.

Definition

A polynomial $p(x) \in F[x]$ is irreducible if and only if $p(x)$ is not constant and has no nontrivial factors.

So $(p(x))$ is maximal if and only if $p(x)$ is irreducible, and $(p(x))$ is prime if and only if $p(x)$ is irreducible or 0.

Note

Two different polynomials can represent the same function. For instance, x^3 and x represent the same function in $\mathbb{Z}_3[x]$ but are different polynomials.

Now, we go on to some things about modules.

Definition

An R -module homomorphism between R -modules M and N is a function $\phi : M \rightarrow N$ with (for $r \in R$, $m, m_1, m_2 \in M$)

- (1) $\phi(rm) = r\phi(m)$
- (2) $\phi(m_1 + m_2) = \phi(m_1) + \phi(m_2)$

Example

If F is a field, an F -module homomorphism is exactly an F -linear map.

Example

Take $R = \mathbb{Z}$, then $\phi : \mathbb{Z}_{12} \rightarrow \mathbb{Z}_3$ given by $\phi(n) = n \pmod{3}$ is a R -module homomorphism.

Example

$\phi : \mathbb{Z}^2 \rightarrow \mathbb{Z}^2$ given by $\phi(a, b) = (a + b, a - b)$ is a $\mathbb{Z}$ -module homomorphism. This is injective but not surjective, which shows how module homomorphisms are not as nice as vector spaces.

Theorem

If $\phi : M \rightarrow N$ is an R -module homomorphism, then $\ker \phi$ and $\text{im } \phi$ are R -submodules of their respective modules.

Proof:

We want to show that $\ker \phi$ is closed under $+$, $-$, $\cdot$ and isn't empty, but $0 \in \ker \phi$. Say $m_1, m_2 \in \ker \phi$. Then $\phi(m_1 \pm m_2) = \phi(m_1) \pm \phi(m_2) = 0 \pm 0 = 0$, so $m_1 \pm m_2 \in \ker \phi$. If $m \in \ker \phi$, and $r \in R$, then $\phi(rm) = r\phi(m) = 0$ so $rm \in \ker \phi$. Similarly, we have $\phi(0) = 0$ so $0 \in \text{im } \phi$. If $n_1, n_2 \in \text{im } \phi$, then $n_1 = \phi(m_1)$, $n_2 = \phi(m_2)$ for some $m_1, m_2 \in M$, so $n_1 \pm n_2 = \phi(m_1 \pm m_2)$ and if $r \in R$, $n \in \text{im } \phi$ with $n = \phi(m)$ so $rn = \phi(rm)$. ■

Question: Is every R -submodule of M the kernel of a homomorphism?

Answer:

Yes. This involves quotients. Again. But every R -module contains an abelian group with other stuff, and we know how to take quotients of abelian groups (any time we have a subgroup, and thus a normal subgroup).

Let $N \subseteq M$ be an R -submodule. We want to define M/N of an R -module. First, define the set M/N and the $+$, $-$ using the quotient of abelian groups. The last step is to define multiplication by elements of R . Let's try $r(m + N) = rm + N$. If $m + N = m' + N$, then $rm - rm' = r \underbrace{(m - m')}_{\in N} \in N$

because N is an R -module. So $rm + N = rm' + N$, so our above multiplication is well-defined. It is now straightforward to show that M/N is an R -module with this multiplication (we get $+$ and $-$ for free by the construction of the quotient).

Theorem (Universal Property of Quotients):

Let $\phi : M \rightarrow L$ be an R -module homomorphism. Let $\pi : M \rightarrow M/N$ be the reduction mod N homomorphism. Then there exists $\tilde{\phi} : M/N \rightarrow L$ such that $\phi = \tilde{\phi} \circ \pi$ if and only if $N \subseteq \ker \phi$. Moreover, $\text{im } \phi = \text{im } \tilde{\phi}$, and $\ker \tilde{\phi} = \pi(\ker \phi)$.

Proof:

The forward direction is easy, since if $\tilde{\phi}$ exists and $\phi = \tilde{\phi} \circ \pi$, then $N = \ker \pi \subseteq \ker \phi$. In the forward direction, define $\tilde{\phi} : M/N \rightarrow L$ by $\tilde{\phi}(m+N) = \phi(m)$. Checking this is well defined, say $m+N = m'+N$. Then $\tilde{\phi}(m+N) = \phi(m)$, $\tilde{\phi}(m'+N) = \phi(m')$ and $m-m' \in N \subseteq \ker \phi$, implying $\phi(m-m') = 0$ implying $\phi(m) = \phi(m')$. We get that $\tilde{\phi}$ is a homomorphism from ϕ . The first moreover follows from the fact that π is surjective, so $\text{im } \phi = \text{im } \tilde{\phi}$. The second moreover holds because $m \in \ker \phi$ if and only if $\phi(m) = 0$ if and only if $\tilde{\phi}(\pi(m)) = 0$ if and only if $\pi(m) \in \ker \tilde{\phi}$. ■

As a warning, we can have ring homomorphisms between R -modules (if it turns out our modules are rings) that aren't R -module homomorphisms:

Example

$\mathbb{C}$ is a $\mathbb{C}$ -module and a ring. Define $\phi(a+bi) = a-bi$. This is a ring homomorphism but not a $\mathbb{C}$ -module homomorphism.

Definition

Let M be an R -module and $S \subseteq M$ a subset. The R -module generated by S is $\{r_1s_1 + \cdots + r_ns_n \mid s_i \in S, r_i \in R\}$, also sometimes called the R -span of S . If $S = \{s_1, \dots, s_r\}$ is finite, then we sometimes denote this by $Rs_1 + \cdots + Rs_r$.

Example

If R is a field, then the module generated by S is literally the span of S .

Example

If $R = \mathbb{Z}$, $M = \mathbb{Z}/20\mathbb{Z}$, $S = \{4, 5\}$. Then the submodule generated by S is $\{4a + 5b \pmod{20} \mid a, b \in \mathbb{Z}\} = \mathbb{Z}/20\mathbb{Z}$ because 4 and 5 are relatively prime.

Example

If $M \subseteq R$, then M is an ideal of R and the R -submodule generated by S is just the ideal generated by S .

Definition

Let S be a set. The free R -module on S is the set of all formal finite R -linear combinations of elements of S (modulo the equivalence relation that makes these combinations actually linear like $r_1s + r_2s = (r_1+r_2)s$ with $(r_1s_1 + \cdots + r_ns_n) + (r'_1s'_1 + \cdots + r'_ms'_m) = r_1s_1 + \cdots + r_ns_n + r'_1s'_1 + \cdots + r'_ms'_m$, and $(r_1s_1 + \cdots + r_ns_n) - (r'_1s'_1 + \cdots + r'_ms'_m) = r_1s_1 + \cdots + r_ns_n + (-r'_1)s'_1 + \cdots + (-r'_m)s'_m$, and $r(r_1s_1 + \cdots + r_ns_n) = (rr_1)s_1 + \cdots + (rr_n)s_n$.

Example

The free $\mathbb{Z}$ -module on $\{\text{John}, \text{Paul}, \text{George}, \text{Ringo}\}$ is $\{a(\text{John}) + b(\text{Paul}) + c(\text{George}) + d(\text{Ringo}) \mid a, b, c, d \in \mathbb{Z}\}$.

Let M be an R -module, and say M is generated by S . Let F be the free R -module on S . There is a homomorphism $\phi : F \rightarrow M$ given by $\phi(r_1s_1 + \cdots + r_ns_n) = r_1s_1 + \cdots + r_ns_n \in M$. Because S generates M , ϕ is surjective. Thus, ϕ induces an isomorphism $\tilde{\phi} : F/\ker \phi \rightarrow M$.

Example

Let $M = \mathbb{Z}/20\mathbb{Z}$, $R = \mathbb{Z}$, $S = \{4, 5\}$. Then $F = \{a(4) + b(5) \mid a, b \in \mathbb{Z}\} \cong \mathbb{Z}^2$. Then $\phi(a(4) + b(5)) =$

$4a + 5b \pmod{20}$. ϕ is surjective, what is $\ker \phi$? We see $\ker \phi$ contains $5(4)$ and $4(5)$, which we claim generates $\ker \phi$. Let $N = \mathbb{Z}[5(4)] + \mathbb{Z}[4(5)]$. Then, mod N , every element of F is congruent to $x(4) + y(5)$ for $x \in \{0, 1, 2, 3, 4\}$ and $y \in \{0, 1, 2, 3\}$. So F/N has at most 20 elements, and it has at least 20 elements since ϕ surjects onto $\mathbb{Z}/20\mathbb{Z}$, so $N = \ker \phi$, since $\ker \phi$ contains N .

Example

Recall $\mathbb{Z}[\zeta_5] = \{a + b\zeta_5 + c\zeta_5^2 + d\zeta_5^3 + e\zeta_5^4 \mid a, b, c, d, e \in \mathbb{Z}\}$. This is the free $\mathbb{Z}$ -module on $\{1, \zeta_5, \zeta_5^2, \zeta_5^3, \zeta_5^4\}$, but as a ring, $\mathbb{Z}[\zeta_5]$ is generated by ζ_5 .

What do abelian groups look like if they're finitely generated? That is, if A is an abelian group, with $A = \langle g_1, \dots, g_n \rangle$, what is A like?

We will find a list of abelian groups such that every finitely generated abelian group is isomorphic to exactly one of the groups on the list. Why are we doing this after modules?:

Fact

Every abelian group is a $\mathbb{Z}$ -module, and vice versa.

Example

Consider $\mathbb{Z}^2$, and X to be the set of all points $(x, y) \in \mathbb{Z}^2$ such that $x \equiv y \pmod{2}$. What's $\mathbb{Z}^2/X$? We have $X = \text{Span}\{(1, 1), (1, -1)\}$. Could choose $\{(1, 1), (1, 0)\} = B$ as a basis of $\mathbb{Z}^2$. Then $X = \text{Span}\{(1, 0)_B, (-1, 2)_B\} = \text{Span}\{(1, 0)_B, (0, 2)_B\}$. Thus reducing $\mathbb{Z}^2$ mod X reduces its first coordinate mod 1 (and becomes 0) and reduces its second coordinate mod 2 (and is either 0 or 1), so $\mathbb{Z}^2/X \cong \mathbb{Z}/2\mathbb{Z}$. The trick was to find the right bases of $\mathbb{Z}^2$ and of X .

We can generalize this:

Lemma

Every subgroup N of $\mathbb{Z}^m$ is free, that is, it's isomorphic to $\mathbb{Z}^t$ for some $t \in \mathbb{Z}_{\geq 0}$.

Proof:

We proceed by induction on m . $m = 0$ is clear, $m = 1$ we have done already. So now consider $M := (\mathbb{Z} \times \underbrace{\{(0, \dots, 0)\}}_{m-1}) \cap N = \{(\star, 0, \dots, 0) \in N\}$. It's a subgroup of $\mathbb{Z} \times \{\mathbf{0}\} \cong \mathbb{Z}$, so it's of the form

$\{(\ell a, 0, \dots, 0) \mid a \in \mathbb{Z}\}$ for some $\ell \in \mathbb{Z}_{\geq 0}$. Let $\phi : \mathbb{Z}^r \rightarrow \mathbb{Z}^{r-1}$ be $\phi(a_1, \dots, a_r) = (a_2, \dots, a_r)$. Then $\phi(N) \subseteq \mathbb{Z}^{r-1}$, so by induction, $\phi(N) \cong \mathbb{Z}^t$ for some t , so $\phi(N) = \mathbb{Z}\mathbf{y}_1 + \dots + \mathbb{Z}\mathbf{y}_t$ for some linearly independent $\mathbf{y}_1, \dots, \mathbf{y}_t \in \mathbb{Z}^{r-1}$. For each i , choose $\mathbf{z}_i \in N$ such that $\phi(\mathbf{z}_i) = \mathbf{y}_i$, so $\mathbf{z}_i = (\star, \mathbf{y}_i)$. These $\mathbf{z}_i$'s are linearly independent because we are adding a coordinate to the already linearly independent set. For any $\mathbf{v} \in N$, we can write $\mathbf{v}$ as a $\mathbb{Z}$ -linear combination of the $\mathbf{z}_i$, except maybe the first coordinate is wrong. In other words, there are $a_1, \dots, a_t \in \mathbb{Z}$ such that $\mathbf{v} - (a_1\mathbf{z}_1 + \dots + a_t\mathbf{z}_t) = (\star, 0, \dots, 0)$. Therefore, $\mathbf{v} - (a_1\mathbf{z}_1 + \dots + a_t\mathbf{z}_t) \in M$ (it's in N by closure under linear combinations). So $\mathbf{v} - (a_1\mathbf{z}_1 + \dots + a_t\mathbf{z}_t) = n(\ell, 0, \dots, 0)$. If $\ell = 0$, then this shows $\{\mathbf{z}_1, \dots, \mathbf{z}_t\}$ is a basis of N . If $\ell \neq 0$, then set $\mathbf{z}_{t+1} = (\ell, 0, \dots, 0)$, and we get a basis $\{\mathbf{z}_1, \dots, \mathbf{z}_{t+1}\}$ of N . So N has a $\mathbb{Z}$ -basis is therefore isomorphic to $\mathbb{Z}^t$ or $\mathbb{Z}^{t+1}$. ■

Theorem (Fundamental Theorem of Finitely Generated Abelian Groups)

Let A be a finitely generated abelian group. Then

$$A \cong \mathbb{Z}^r \times (\mathbb{Z}/p_1^{a_1}\mathbb{Z}) \times (\mathbb{Z}/p_2^{a_2}\mathbb{Z}) \times \dots \times (\mathbb{Z}/p_n^{a_n}\mathbb{Z})$$

where the p_i are prime, $r \geq 0 \in \mathbb{Z}$, and $a_1, \dots, a_m > 0$. Moreover, this expression is unique up to permutation of the $p_i^{a_i}$'s.

Proof:

Say $A = \langle g_1, \dots, g_m \rangle$. Then there is a homomorphism from the free $\mathbb{Z}$ -module: $\phi : \mathbb{Z}^m \rightarrow A$ given by $\phi(x_1, \dots, x_m) = x_1g_1 + \dots + x_mg_m$ that is surjective. So $A \cong \mathbb{Z}^m / \ker \phi$ by the First Isomorphism Theorem. By the lemma, $\ker \phi$ has a basis $\{\mathbf{y}_1, \dots, \mathbf{y}_t\}$. Consider the matrix $Y = [\mathbf{y}_1 \mid \dots \mid \mathbf{y}_t]$. Row operations on Y will rewrite the columns in a different basis of $\mathbb{Z}^m$. Column operations will change the

basis of $\ker \phi$ to a different basis of $\ker \phi$. So performing row and column operations on the matrix Y will not change the isomorphism type of A . So permute to row 1 the row with the smallest-in-absolute-value nonzero entry in the first column. Use row operations to reduce all the other entries in column to be less than the top entry. Repeat this process until all entries in column 1 are zero, except the first one. Then use column operations to do the same for row 1: we want the upper left entry to be the only nonzero entry in the first row. If necessary, go back and fix column 1 again. Repeat these processes in row 1 and column 1 until both are all zero except the upper left entry. This is possible since the upper left entry goes down each time, and we are working with integers, so we can only go down finitely many times. Now repeat on the smaller matrix B obtained by deleting the first row and column. Recursively, we eventually get a diagonal matrix $\text{diag}(d_1, \dots, d_t)$ (where we can have zero rows at the bottom) where the d_i are integers. So now by taking quotients of a diagonal matrix, we see $A \cong (\mathbb{Z}/d_1\mathbb{Z}) \times \dots \times (\mathbb{Z}/d_t\mathbb{Z})$. If $d_i = 0$, then $\mathbb{Z}/d_i\mathbb{Z} = \mathbb{Z}$. So $A \cong \mathbb{Z}^r \times (\mathbb{Z}/e_1\mathbb{Z}) \times \dots \times (\mathbb{Z}/e_k\mathbb{Z}) \cong \mathbb{Z}^r \times (\mathbb{Z}/p_1^{a_1}\mathbb{Z}) \times \dots \times (\mathbb{Z}/p_n^{a_n}\mathbb{Z})$ by the Chinese remainder theorem. This proves existence.

Proving uniqueness, define $T \subseteq A$ to be the set of all elements in A of finite order, where T is exactly the copy of $(\mathbb{Z}/p_1^{a_1}\mathbb{Z}) \times (\mathbb{Z}/p_b^{a_b}\mathbb{Z})$ inside A . If $A \cong A'$, then $A/T \cong A'/T'$. Write $A \cong \mathbb{Z}^r \times T$ and $A' \cong \mathbb{Z}^{r'} \times T'$, implying $\mathbb{Z}^r \cong A/T \cong A'/T' \cong \mathbb{Z}^{r'}$ so $r = r'$. Define $T_p = \{g \in A \mid |g| = p^n \text{ for some } n\}$, and similarly for T'_p and A' . It is enough to show T_p and T'_p have the same representation, since $T \cong \prod_p T_p$. Let n_k be the number of elements of A whose order is p^k , and same for A' . We know $n_1 = p^b$ and $n'_1 = p^{b'}$, where b and b' are the number of factors where $p_i = p$, but since $T_p \cong T'_{p'}$, we have $p^n = p^{n'}$, implying $n = n'$. Similarly, we have $n_2 = (p^2)^{b_2} p^{b-b_2} = (p^2)^{b'_2} p^{b-b'_2}$ where b_2 is the number of factors where $p^2 \mid p_i^{a_i}$ (and analogously for b'_2), getting that $b_2 + b = b'_2 + b \implies b_2 = b'_2$. Inductively, we have $n_k = (p^k)^{b_k} P_{k-1} = (p^k)^{b'_k} P_{k-1}$, where P_{k-1} is the product of terms from the $k-1$ step, implying $b_k = b'_k$. Therefore, by induction, we see the two representations are identical. ■

Say D is a domain. Then if $ab = cb$, $b \neq 0$. This implies $a = c$, because $b(a - c) = 0$, implying $a - c = 0$. Let's develop the technology to just divide both sides by b .

We want to add an element " $1/b$ ". But the only way we know to make a bigger ring is to adjoin a variable to it, say $D[x]$. But then to make b a unit, we can kill off $bx - 1$, to get $D' = D[x]/(bx - 1)$. Then in D' , $bx = 1$, so b is a unit. We will see later that D' is a domain that contains D (there is an injective homomorphism from D to D' , it's injective because no constant can divide the degree 1 polynomial $bx - 1$).

Alternatively, you could do this. Let $K(D) = \{a/b \mid a, b \in D, b \neq 0\} / \sim$ where $\frac{a}{b} \sim \frac{c}{d}$ if and only if $ad = bc$, where $\frac{a}{b} \pm \frac{c}{d} = \frac{ad \pm bc}{bd}$ and $\frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}$. These operations make $K(D)$ into a ring, called the fraction field of D where $1 = \frac{1}{1}$. Also, there is an injective homomorphism $\phi: D \hookrightarrow K(D)$ by $\phi(d) = \frac{d}{1}$, where often we write d instead of $\frac{d}{1}$.

Example

$K(\mathbb{Z}) = \mathbb{Q}$.

Example

$K(\mathbb{R}[x]) = \{\frac{p(x)}{q(x)} \mid p(x), q(x) \in \mathbb{R}[x], q(x) \neq 0\}$ is the rational functions in x , denoted by $\mathbb{R}(x)$.

Example

$K(\mathbb{C}[x, y, z]) = \mathbb{C}(x, y, z)$ is the rational functions in three variables over $\mathbb{C}$.

Theorem

$K(D)$ is a field.

Proof:

The inverse of an element a/b is b/a , as long as $a \neq 0$. ■

Theorem

Let D be a domain and F be a field, and say $\phi : D \rightarrow F$ is an injective homomorphism. Then there is an injective homomorphism $\psi : K(D) \rightarrow F$ satisfying $\phi = \psi \circ i$, where $i(d) = \frac{d}{1}$ is the inclusion homomorphism.

Proof:

Define $\psi : K(D) \rightarrow F$ by $\psi(\frac{a}{b}) = \frac{\phi(a)}{\phi(b)}$. Checking this is well-defined, if $\frac{a}{b} = \frac{c}{d}$, then $ad = bc$, so $\phi(a)\phi(b) = \phi(b)\phi(c)$ so $\frac{\phi(a)}{\phi(b)} = \frac{\phi(c)}{\phi(d)}$. Moreover, $\phi(b) = 0$ only happens when $b = 0$ because ϕ is injective.

ψ is also a straightforwardly a homomorphism. It is injective because $K(D)$ is a field (using a previously proved property). Finally, $(\psi \circ i)(d) = \psi(\frac{d}{1}) = \frac{\phi(d)}{\phi(1)} = \phi(d)$. ■

Now, if D is a domain, $b \in D$ a nonzero element, then $D[1/b] \subseteq K(D)$ is a ring (domain) in which b is a unit. In fact, you can do this for arbitrary subsets of $D \setminus \{0\}$.

Definition

Let D be a domain. A multiplicative subset of D is a subset $S \subseteq D \setminus \{0\}$ that is closed under multiplication.

Definition

The localization of D at a multiplicative subset S is $D_S = \{\frac{a}{b} \mid a \in D, b \in S\}$. D_S is a subring of $K(D)$.

Example

Let D be a domain, $P \subseteq D$ a prime ideal. The localization of D “at P ” is $D_P = \{\frac{a}{b} \mid a \in D, b \notin P\}$. This is actually the localization of D at $D \setminus P$, but no one ever says that. This construction requires P to be prime, because otherwise $D \setminus P$ is not a multiplicative set.

Definition

Let I and J be ideals of a ring R . Then $I + J = \{a + b \mid a \in I, b \in J\}$, and IJ is the ideal generated by $\{ab \mid a \in I, b \in J\}$.

Theorem (Chinese Remainder)

Let R be a ring, and let $I, J \subseteq R$ be ideals of R , with $I + J = R$. Then

$$R/IJ \cong (R/I) \times (R/J)$$

Proof:

Define $\phi : R \rightarrow (R/I) \times (R/J)$ by $\phi(r) = (r + I, r + J)$. The kernel of ϕ is $\ker \phi = I \cap J$. Since $IJ \subseteq I \cap J$, we get a homomorphism $\tilde{\phi} : R/IJ \rightarrow (R/I) \times (R/J)$ by the Universal Property of Quotients. $\tilde{\phi}$ is surjective because $I + J = R$, so there are $a \in I, b \in J$ with $a + b = 1$ (this is the Euclidean algorithm essentially). So for any $(x, y) \in (R/I) \times (R/J)$, we have $\phi(bx + ay) = (x, y)$. The kernel of $\tilde{\phi}$ is $\ker \phi = I \cap J$ reduced mod IJ , and $I \cap J = (I \cap J)R = (I \cap J)(I + J) = (I \cap J)I + (I \cap J)J \subseteq IJ + IJ = IJ$. Since know $IJ \subseteq I \cap J$, this gives $I \cap J = IJ$, so $\tilde{\phi} : R/IJ \rightarrow (R/I) \times (R/J)$ is an isomorphism. ■

Example

$\mathbb{R}[x]/(x^2 + 3x + 2) = \mathbb{R}[x]/((x + 1)(x + 2)) \cong \mathbb{R}[x]/(x + 1) \times \mathbb{R}[x]/(x + 2) \cong \mathbb{R} \times \mathbb{R} = \mathbb{R}^2$.